

Lithium-ion battery degradation modelling using universal differential equations: Development of a cost-effective parameterisation methodology

Jishnu Ayyangatu Kuzhiyil^{a,b,*}, Theodoros Damoulas^{c,d}, Ferran Brosa Planella^{b,e},
W. Dhammika Widanage^a

^a WMG, University of Warwick, Coventry, UK

^b The Faraday Institution, Quad One, Harwell Science and Innovation Campus, Didcot, UK

^c Department of Computer Science and Department of Statistics, University of Warwick, Coventry, UK

^d The Alan Turing Institute, London, UK

^e Mathematics Institute, University of Warwick, Gibbet Hill Road, Coventry, CV4 7AL, UK

ARTICLE INFO

Dataset link: <https://github.com/JishnuKuzhiyil/Calendar-Ageing-Modelling-using-UDE---Julia-Code.git>, <https://zenodo.org/records/14577286>

Keywords:

Universal differential equations
Scientific machine learning
Battery degradation modelling
Calendar ageing

ABSTRACT

The accuracy and reliability of physics-based lithium-ion battery degradation models are limited by incomplete understanding of degradation mechanisms. This article presents Universal Differential Equations (UDE) based degradation modelling, which integrates neural networks into physics-based model differential equations to learn partially understood degradation mechanisms. Therefore, this approach combines the function approximation capabilities of machine learning with the interpretability of physics-based models. However, the widespread adoption of this methodology is hindered by the high cost of training neural networks placed within a physics-based degradation model. To address this, we propose a cost-effective parameterisation method that exploits the large difference between electrochemical and degradation time scales, to speed up the gradient calculation using the continuous adjoint sensitivity analysis. Additionally, efficient scaling of this method to multiple ageing datasets is ensured through mini-batching. Finally, we demonstrate this approach by developing a novel UDE calendar ageing model and validating it against in-house experimental data covering 39 storage conditions (13 states of charge at 0 °C, 25 °C, and 45 °C). The predictions on full cell capacity and loss of active material (LAM) at negative electrode align well with experimental observations with an average RMSE of 0.66% and 1.11% respectively, which was a significant improvement over the baseline physics-based model.

1. Introduction

Lithium-ion batteries (LIBs) are important in humanity's progress towards a sustainable, low-carbon future. Mathematical models of LIBs help engineers to create optimal designs for battery-powered systems, and also to make informed decisions about their safe and efficient operation. For example, during the design phase of an electric aircraft, model-based optimisation is used to make informed choices on the design of its battery pack, thermal management system, charging protocols etc.

Among various LIB modelling approaches, continuum physics-based models stand out, due to the high degree of physical insight they offer [1]. Continuum models consist of differential algebraic equations that describe the conservation and transport laws of charge and mass species within the battery [2]. Once formulated and parameterised, these models can be integrated with platforms like PyBaMM [3] to

run virtual experiments that simulate real-world battery operations and predict corresponding battery responses.

Continuum battery modelling starts with an electrochemical model, such as the Doyle–Fuller–Newman (DFN) model, which describes the transport of lithium within the battery [4]. To simulate battery degradation, additional sub-models that represent relevant degradation mechanisms must be coupled with the electrochemical model [5,6]. The accuracy of long-term simulations relies on how accurately these degradation mechanisms are represented. However, battery degradation mechanisms are not yet fully understood and remain an active area of research [7,8]. This incomplete understanding of the degradation physics could compromise the accuracy of the physics-based degradation models and of any subsequent decisions made about the safe and efficient operation of batteries [9].

The traditional approach to addressing this challenge involves conducting more scientific studies to understand the missing physics

* Corresponding author at: WMG, University of Warwick, Coventry, UK.
E-mail address: jishnu-ak.ayyangatu-kuzhiyil@warwick.ac.uk (J.A. Kuzhiyil).

and subsequently incorporating them into the continuum models. However, this method is prohibitively time-consuming and expensive. Moreover, due to the complex nature of battery degradation mechanisms, even extensive research may not guarantee a complete and correct model of all the relevant physics. The emerging field of scientific machine learning (SciML) offers an alternative solution [10,11].

SciML is a modelling philosophy that combines mechanistic information from a physics-based model (i.e., causation) with correlations from data to account for unknown physics [12,13]. In particular, the SciML framework of Universal Differential Equations (UDEs) allows the integration of neural networks within the state equations of a physics-based model [13,14]. The UDE approach can be used to improve continuum degradation modelling by using neural networks to enhance sub-models describing poorly understood degradation mechanisms. Within this setting, the learning of missing physics becomes equivalent to the parameterisation of the UDE model.

Although articles [15] and [16] have used UDE for battery degradation modelling, they have not fully addressed the challenge of improving models that describe degradation mechanisms using measurable data. For example, [15] did not use neural networks to improve any degradation mechanism, but instead used them to model the residual of aggregate quantities such as capacity and resistance, which the physics-based model failed to capture. Although this formulation improved the accuracy of capacity and resistance predictions, it did not improve the accuracy of other important outputs, such as degradation modes. On the other hand, [16] demonstrated the ability of the UDE framework to model the growth of solid electrolyte interface (SEI) during cell formation. However, they considered synthetic data of internal battery variables generated using a physics-based model, which are not measurable in practice.

This article aims to show how the UDE framework can be used to improve submodels that describe degradation mechanisms within a continuum electrochemical degradation model using real, measurable data. We identified the high cost of parameterising the UDE degradation model as a major roadblock to achieving this goal. The parameterisation is performed as an optimisation problem, which requires numerous forward runs of the UDE degradation model. However, each forward run is expensive due to the coupling between the electrochemical and degradation models [4]. The coupling constrains the numerical solver to take time steps on the time scale of electrochemical dynamics (spanning seconds to minutes), while the ageing dynamics span over months. Thus, coupling makes the overall optimisation process highly expensive (see Section 3.1 for a detailed explanation). To advance UDE degradation research, it is imperative to develop a faster and more efficient parameterisation methodology.

To this extent, this research article develops a novel approach suitable for parameterising UDE degradation models. The methodology comprises two key components. First, inspired by the work of Sulzer et al. [17], we created a methodology that facilitates faster gradient calculation by exploiting wide differences in electrochemical and degradation time scales. This method is suitable when a single ageing dataset is considered. The second component involves mini-batching techniques [18], which enable efficient scaling of the optimisation when multiple ageing datasets are present. The combination of these two strategies allows faster and cheaper parameterisation of UDE degradation models.

This article demonstrates the formulation and parameterisation of a UDE degradation model that explains calendar ageing. Calendar ageing refers to battery degradation under zero load conditions [19,20]. Battery-powered devices, including electric vehicles and most consumer electronics, spend $\geq 90\%$ of their lifetime in an idle state, which emphasises the importance of developing accurate calendar ageing models [21]. The research used an in-house collected calendar ageing data set, hereafter referred to as the WMG-calendar ageing data set. The ageing experiment involved storing LGM-50T commercial cells, at 0, 25 and 45 °C, and 13 different State of Charge (SOC) points per

temperature.

The literature reports that continual growth of the Solid Electrolyte Interphase (SEI) layer is the dominant mechanism causing calendar ageing [22,23]. We show that commonly used models for SEI growth, particularly the Tafel kinetics limited by the solvent diffusion [4,24], fail to accurately explain the trends observed in our experimental data. To rectify this, we introduce a UDE degradation model and parameterised it with our newly developed parameterisation method. The major contributions of the article are described below.

1.1. Research contributions

1. This article addresses the high computational cost of parameterising the UDE degradation model. The developed methodology exploits the wide difference in model time scales to accelerate the gradient calculation step involved during parameter optimisation. Furthermore, the mini-batching technique is used to efficiently scale the method when multiple ageing datasets are involved.
2. The limitations of the physics-based SEI growth model are shown by validating it against an extensive calendar ageing dataset, comprising of three storage temperatures and 13 storage SOCs per temperature. To the best of our knowledge, this is the first work to evaluate SEI models across such a wide range of ageing conditions.
3. This is the first UDE-based model for calendar ageing with ML models integrated within the state equations of a continuum physics-based degradation model. The developed UDE model outperformed the widely used physics-based models in accuracy across most storage conditions, including those not used for UDE training. On average, the RMSE of capacity and anode LAM predictions by the UDE model was 58.72% and 33.0% lower, respectively, compared to the physics-based model.

1.2. Organisation of the article

Section 2 systematically introduces the UDE degradation modelling framework. Section 3 explains the challenges associated with the parameterisation of the UDE degradation models. Additionally, Section 3 also explains the development of Accelerated Parameterisation via Timescale Decoupling (APTD) and mini-batching approaches to address the parameterisation challenges. Section 4 presents the calendar ageing dataset and develops physics-based and UDE degradation models for battery calendar ageing. Section 5 describes the parameterisation and validation of both models. Section 6 compares the presented calendar ageing modelling approach against traditional approaches. Finally, the conclusions of the study are presented in Section 7.

2. UDE degradation modelling

Physics-based degradation models are a combination of an electrochemical model coupled with additional differential equations describing the degradation mechanisms. To simulate long-term battery behaviour, both models must be solved simultaneously [1]. Let $\mathbf{x} \in \mathbb{R}^{n_1}$ and $\mathbf{z} \in \mathbb{R}^{n_2}$ denote the electrochemical and degradation state variables, respectively. The coupled system can be denoted in state space form as

$$\text{Electrochemical dynamics: } \frac{d\mathbf{x}}{dt} = f(\mathbf{x}, \mathbf{z}, \mathbf{u}, t; \boldsymbol{\alpha}), \quad (1a)$$

$$\text{Degradation dynamics: } \frac{d\mathbf{z}}{dt} = g(\mathbf{z}, \mathbf{x}, \mathbf{u}, t; \boldsymbol{\theta}), \quad (1b)$$

where $\mathbf{u}(t) \in \mathbb{R}^m$ denotes the external forcing term (typically load current in the context of batteries), and $\boldsymbol{\alpha}$ and $\boldsymbol{\theta}$ are the parameter vectors. In this research, we consider that the electrochemical model in Eq. (1a) and the parameter vector $\boldsymbol{\alpha}$ is known. This is a fair assumption, as begin-of-life (BoL) parameterisation is performed prior to any ageing

model development [4]. The evolution of electrochemical dynamics with ageing is captured by updating the degradation state z in Eq. (1a).

However, the degradation state function $g(\cdot)$ in Eq. (1b) is only partially known, and UDE degradation modelling aims to identify the unaccounted dynamics from experimental data. To achieve this, the state function $g(\cdot)$ in Eq. (1b) is replaced with a new state function $h(\cdot)$. The function $h(\cdot)$ is constructed such that the known physics is retained, and the functions representing the unknown physics are replaced by feed-forward neural networks $\mathcal{N}_i(\cdot)$, allowing them to learn the missing physics. Thus, the UDE degradation model is expressed in the following general form

$$\frac{dx}{dt} = f(x, z, u, t; \alpha), \quad (2a)$$

$$\frac{dz}{dt} = h(z, \mu(x, u), t, \mathcal{N}_1, \mathcal{N}_2, \dots, \mathcal{N}_N; \theta). \quad (2b)$$

The function $\mu(x, u) : \mathbb{R}^{n_1} \times \mathbb{R}^m \rightarrow \mathbb{R}^q$ is introduced to emphasise that degradation dynamics typically depend on a lower-dimensional mapping of the electrochemical states and external forcing terms (generally $q \leq n_1 + m$). This observation helps to reduce the memory requirement for the parameterisation method developed in this article. The neural networks $\mathcal{N}_i(z, \mu(x, u), t)$ can take the same inputs as the state function h . Furthermore, the parameter vector θ also includes neural network weights and biases.

The formulation of the function $h(\cdot)$ is a crucial step in UDE degradation modelling. The modeller must make decisions such as how to retain the known physics, where to embed neural networks, and how to select their architecture, inputs, and outputs based on domain knowledge. As UDE degradation modelling research is in its early phase, standardised guidelines for designing the UDE degradation state function $h(\cdot)$ do not exist. Progress in this field requires researchers to address individual degradation modelling problems and to progressively establish standardised guidelines and practices. As a step in this direction, this article performs calendar ageing modelling through the UDE framework. The process of constructing the UDE degradation state function for calendar ageing is explained in Section 4.4.

3. Parameterisation of the UDE degradation model

3.1. Overview of challenges

Parameterisation of the UDE degradation model involves the estimation of θ in (2b), by optimising a loss function $\mathcal{L}$ that penalises the error between the predictions of the model and the experimental observations. Note that, it is during parameterisation, the neural networks learn the missing physics. Since the vector θ includes parameters of the neural networks, it is preferable to use gradient-based optimisers [25]. These optimisers require the calculation of the gradient $\frac{d\mathcal{L}}{d\theta}$, which in turn depends on the gradient of the solution of the UDE degradation model with respect to its parameters [13,26].

Continuous Adjoint Sensitivity (CAS) is the recommended method for calculating the gradient of differential equation solutions to its parameters, especially when a large number of parameters are present [27]. The gradient calculation using CAS involves solving the original differential equations in the forward time direction, followed by solving adjoint differential equations in the reverse time direction [13,26]. The solution of the adjoint differential equations enables the calculation of the required gradient, refer to Appendix A for a summary of adjoint calculations.

The parameterisation of UDE degradation models is challenging for two reasons. First, as a large number of parameters are involved, many iterations are needed during the optimisation. Second, the gradient calculation in each iterations is very time consuming. This is because CAS requires forward and inverse solving of a system of differential equations with $n_1 + n_2$ states for an extended simulation period (typically spanning months), while the adaptive solver is limited to taking time

steps in the electrochemical time-scale range (seconds to minutes). The complexity further increases when the loss function involves multiple ageing datasets, as each iteration requires solving these equations for multiple conditions.

To rectify the challenges mentioned above, a novel parameterisation method is developed, consisting of two methods: Accelerated Parameterisation via Timescale Decoupling (APTD) and mini-batching. APTD allows for faster gradient calculation when a single ageing dataset is involved. Further, mini-batching enables efficient scaling of APTD when multiple ageing datasets are involved. Details of these methods are provided in Sections 3.2 and 3.3.

3.2. Accelerated parameterisation via timescale decoupling (APTD)

In this Section, we discuss the development of the APTD methodology, which enables faster parameterisation of UDE degradation models by accelerating the gradient calculation step. The APTD methodology is strongly inspired by Sulzer et al.'s adaptive inter-cycle acceleration algorithm [17]. Their algorithm exploits the substantial time-scale difference between electrochemical and degradation dynamics to speed up physics-based degradation model simulations, by about two orders of magnitude. Building on [17], Pannala et al. developed a faster parameterisation methodology for physics-based degradation models using a derivative-free optimiser [28]. However, derivative-free optimisers are not suitable for UDE degradation models, due to their large number of parameters.

The APTD methodology presented here extends Pannala's methodology to make it suitable for UDE degradation modelling. As mentioned in Section 3.1, slowness of gradient calculation arises from the solver's limitation in taking smaller time steps for extended simulations, a consequence of the coupling between electrochemical and degradation dynamics. APTD reformulates the coupled system in Eq. (2) into a new system of equations containing only the degradation states, allowing the numerical solver to take larger time steps, enabling faster gradient calculation.

The APTD methodology comprises the following three steps.

1. Estimation of the degradation states $z(t)$ from the ageing dataset.
2. Construction of the augmented dataset.
3. Reformulation of the degradation differential equation (Eq. (2b)) for faster gradient calculation.

The first step involves estimating $z(t)$ using an ageing dataset and the known electrochemical model in Eq. (2a). Note that, degradation state variables often show linear, self-limiting or accelerating trends over time, which can be approximated using lower-order polynomials [5,17]. Keeping this in mind, two approaches can be introduced.

Approach 1: Estimate z at discrete time points by performing multiple independent parameter estimations on Eq. (2a), using dataset over short time intervals where z remains fairly constant. Then, approximate the continuous $z(t)$ with a lower-order polynomial. This approach is suitable for state variables, like the active material volume fraction, that can be estimated through degradation mode quantification [29, 30].

Approach 2: Assume a parametric function for $z(t)$ with very few parameters based on prior knowledge and substitute it in Eq. (2a). Then estimate its parameters using ageing data. For example, SEI layer thickness shows a self-limiting trend [4] and can be approximated using a logistic function or square root function. This approach is more suitable for degradation state variables which cannot be directly estimated from degradation mode quantification, such as SEI layer thickness.

In practice, the modeller can employ both approaches, choosing the appropriate one for each element of z . Furthermore, modellers may also explore other approaches but the main goal remains estimating $z(t)$ for ageing data. The estimated value is denoted as $\hat{z}(t)$.

After estimating $\hat{z}(t)$ from the ageing dataset, the next step involves constructing the augmented dataset. To do so, we substitute $\hat{z}(t)$ and $u(t)$ from the ageing data set into the electrochemical model in Eq. (2a) and simulate it to obtain estimates of electrochemical states, denoted as $\hat{x}(t)$. Using $\hat{x}(t)$ and $u(t)$, calculate $\mu(t) = \mu(\hat{x}(t), u(t))$ for the ageing dataset. Subsequently, an augmented dataset $\mathcal{D} = \{t, \hat{z}(t), \mu(t)\}$ can be constructed and saved. To reduce memory requirements for storing the augmented dataset, we store only key data characteristics instead of the entire time series. For example, for constant current cycling ageing, if $\mu(t)$ contains input current, we store only the discharge/charge current values and duration of each step.

After constructing the augmented dataset $\mathcal{D}$, the next step is to reformulate the model for faster gradient calculation. To begin, divide the total simulation time into multiple disjoint time intervals $\mathcal{T}_i = [t_i^-, t_i^+]$, where t_i^- and t_i^+ are the start and end times of the i th segment. The duration of the segments need not be identical but should be selected so that z remains fairly constant within each segment. Further, a new time variable $\tau := t/\gamma$, where $\gamma \gg 1$, is chosen such that on average one unit of τ roughly equals the duration of a segment. Furthermore, we introduce the function $\sigma(\tau)$ which maps τ to the time segment $\mathcal{T}_i$ such that $t = \gamma\tau \in \mathcal{T}_i$.

To this end, the degradation UDE equation for the ageing data considered can be reformulated as

$$\frac{dz}{d\tau} = \left(\frac{\gamma}{\sigma(\tau)} \right) \cdot \int_{\sigma(\tau)} h(z, \mu, t, \mathcal{N}_1, \mathcal{N}_2, \dots, \mathcal{N}_N; \theta) dt. \quad (3)$$

Eq. (3) can be interpreted as the rate of change of z in the degradation time scale τ equals the average rate of change over a short time interval around τ . Since $\mu(t)$ is pre-calculated and stored in the augmented dataset, it acts as a known external input when solving Eq. (3). This means Eq. (3) can be solved independently of the electrochemical model, allowing the solver to take large time steps, making gradient calculation using CAS faster.

Further, the loss function can be formulated as,

$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^N \|z_i - \hat{z}_i\|_2^2, \quad (4)$$

where the $\hat{z}_i$ is the estimated degradation states and z_i is the solution to Eq. (3) at i th time point. An important feature of loss function $\mathcal{L}$ is its differentiability, as it is formulated directly based on difference between solution of a differential equation (z) and data ($\hat{z}$). Thus, CAS method can be directly used to find the gradient (refer to Appendix A or [26]).

On the other hand, formulating the loss function based on capacity measurement would make the gradient calculation more challenging. This is because, in model based simulations, capacity is typically calculated as the time interval between two discrete events: when the battery voltage reaches its upper and lower cut-off values. Calculating the derivative of time difference between discrete events is not straightforward. This complexity explains why Pannala et al. [28] used derivative-free optimisers in their work, where they used a loss function based on capacity measurement.

Decoupling electrochemical and degradation dynamics during parameterisation, as implemented in APTD, offers another key advantage. It prevents the electrochemical model from crashing during optimisation, when the optimiser suggests parameter values that result in physically infeasible degradation states [31]. This makes the optimisation smoother and more robust.

In summary, the APTD method reformulates the gradient calculation to enhance its speed, ensures differentiability of the loss function, and makes the optimisation process smoother.¹

¹ APTD is applicable to both cyclic and calendar ageing model parameterisation. During calendar ageing, as electrochemical dynamics is inactive, ODE solvers can use larger time steps even without APTD problem reformulation. However, APTD problem reformulation is still useful in calendar ageing as it create a differentiable loss function.

3.3. Mini-batching

When multiple ageing datasets are available, with cells aged under different conditions, the conventional approach involves formulating the full batch loss:

$$\mathcal{L}_{FB} = \frac{1}{2} \sum_{j=1}^{N_D} \sum_{i=1}^{N_j} \|z_{i,j} - \hat{z}_{i,j}\|_2^2, \quad (5)$$

where N_D represents the number of ageing datasets, and for each j th dataset, there are N_j data points. Optimisation of full batch loss in Eq. (5) requires the calculation of $\frac{d\mathcal{L}_{FB}}{d\theta}$ using the CAS approach during each optimisation iteration. Calculating $\frac{d\mathcal{L}_{FB}}{d\theta}$ involves solving N_D differential equation systems in forward and reverse directions. As the number of datasets (N_D) increases, this computation becomes exceedingly complex. To address this limitation, this study employs the mini-batching approach.

Mini-batching is a widely used machine learning model parameterisation approach, that has lately been applied for parameterising system of differential equations [18]. This section aims to introduce the effectiveness of the mini-batching to physics-based degradation modelling community, which typically relies on full-batch loss functions even when multiple ageing datasets are involved [28]. Although mini-batching is proposed here for parameterising the UDE degradation model, this approach is equally applicable to the parameterisation of physics-based models as well.

Mini-batching involves dividing the entire dataset into smaller mini-batches [14]. During each optimiser iteration, the gradient is calculated using the part of the full batch loss function associated with the data points in that mini-batch. During each iteration, an individual mini-batch is utilised, progressively passing through all the mini-batches in a single full cycle, referred to as an epoch. Multiple epochs are run until the specified termination criterion. This study treats each ageing dataset as a mini-batch, resulting in a total of N_D mini-batches. Since mini-batching only uses one ageing dataset per optimisation iteration, the gradient calculation cost does not increase with the number of datasets considered.

Mini-batching also offers several other advantages over full-batch optimisation. It introduces stochasticity in the gradient calculation, helping the optimiser to avoid sharp local minima and saddle points [32]. Additionally, as reported by [18], the overall parameterisation time for differential equation systems was reduced by order of magnitude when employing mini-batching instead of full-batch optimisation. Moreover, the frequent parameter updates in mini-batching facilitate a more efficient exploration of the parameter space compared to full-batch methods.

3.4. Summary of reduction in complexity

To summarise, the computational complexity of parameterising the UDE degradation model in Eq. (2) with N_D datasets without APTD and mini-batching is

$$\mathcal{O}\left(N_D \times \frac{T}{\Delta t_{\text{elec}}} \times (n_1 + n_2)\right),$$

per optimiser iteration. Here, T is the total simulation time (assumed identical for all datasets) and Δt_{elec} is the solver time step within the electrochemical process time scale. In contrast, the problem reformulation via the APTD method reduces the complexity to

$$\mathcal{O}\left(N_D \times \frac{T}{\Delta t_{\text{deg}}} \times n_2\right),$$

where Δt_{deg} denotes solver time steps within the degradation time scale. The reduction is achieved because the reformulated problem in Eq. (3) involves only n_2 states, instead of $n_1 + n_2$ states in Eq. (2). Furthermore, the solver is allowed to take time steps in the degradation time scale ($\Delta t_{\text{deg}} \gg \Delta t_{\text{elec}}$).

Table 1

The duration of storage for cells undergoing calendar ageing under different storage conditions. Each row corresponds to a specific storage temperature and each column corresponds to a specific storage SOC.

Storage duration [Days]		Storage SOC (%)												
		0%	2%	5%	10%	30%	50%	60%	70%	80%	85%	90%	95%	100%
Storage Temperature	0 °C	746	746	746	746	746	746	746	746	755	746	755	746	755
	25 °C	740	740	744	744	662	744	662	662	496	740	740	740	496
	45 °C	744	744	744	662	496	662	496	511	511	770	511	752	496

Combining APTD with mini-batching further lowers the complexity per iteration to

$$\mathcal{O}\left(\frac{T}{\Delta t_{\text{deg}}} \times n_2\right),$$

as only one of the N_D ODE systems is solved per iteration.

4. Calendar ageing modelling

This article demonstrates a calendar ageing model created using the UDE degradation modelling approach. Calendar ageing refers to the process of battery degradation during storage. Section 4.1 describes the in-house WMG calendar ageing dataset used for the modelling. Further, as discussed in Section 4.2, Differential Voltage Analysis (DVA) was performed to identify the dominant degradation mechanisms causing the calendar ageing. Based on insights from the DVA, an appropriate physics-based model was chosen from the literature, as discussed in Section 4.3. Finally, to address the inadequacies of the physics-based model (discussed in results Section 5.1), a UDE degradation model was developed, as explained in Section 4.4.

4.1. Experimental dataset

The in-house collected WMG calendar ageing dataset consists of 117 LGM-50T commercial cells. Cells were stored at three ambient temperatures (0, 25, and 45 °C) and 13 storage SOC per temperature, with three cells stored under each condition. The storage duration ranges between 496–770 days. Table 1 shows the number of days calendar-aged, the storage SOC, and storage temperature under each condition. The variation in storage duration across different storage experiments was due to practical difficulties associated with long-term ageing experiments, including operational and resource constraints. The LGM-50T cells have an NMC-811 cathode and silicon-graphite composite anode with 2% volume fraction of silicon [33]. See [34,35] for a detailed description of cell properties.

During storage, the cells were kept at open circuit condition inside three different thermal chambers. To check the ageing status, reference performance tests (RPTs) were carried out during different stages of ageing. During the RPTs, the cells were transferred to different thermal chamber maintained at 25 °C, and connected to Digatron cyclers. Detailed explanation of experimental setup and experimental procedure is provided in supplementary material Section S2. The RPTs included capacity measurement at a $C/3$ discharge rate and resistance measurement at different SOC using pulse tests. After each RPT, the SOC definition was updated based on the latest $C/3$ capacity, and cells were brought to the storage SOC according to the updated definition. Some RPTs, named as the RPT extended (RPTx), also included a $C/20$ (dis)charge step, which allows for the quantification of degradation modes.

The capacity fade during calendar ageing is typically attributed to the continuous growth of the SEI layer [23]. The SEI growth is caused by parasitic side reactions that involve the reduction of the electrolyte at the anode. These side reactions lead to a stoichiometric drift between electrodes, resulting in a decrease in cell capacity [29]. Although the SEI side reactions should only cause loss of lithium inventory (LLI), recent studies have also reported loss of active material (LAM) in the

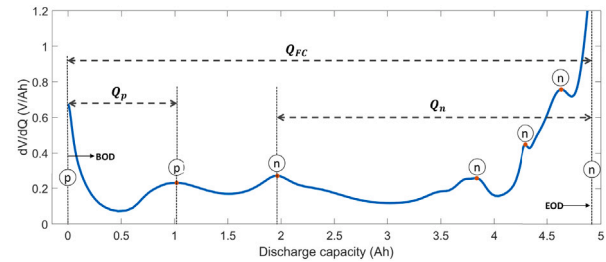


Fig. 1. Differential voltage (DV) plot of a fresh LGM-50T commercial cell generated from constant current discharge data at a $C/20$ current rate. The peaks associated with the negative electrode (anode) are marked with “n” and the peaks associated with the positive electrode (cathode) are marked with “p”. BOD and EOD stand for beginning of discharge and end of discharge, respectively.

electrodes [23,36,37]. Understanding the dominant degradation modes (DMs) is crucial for choosing appropriate degradation differential equations [7]. This study used differential voltage analysis (DVA) to gain insight into degradation modes [38].

Note that, calendar ageing rates are generally expected to increase monotonically with storage SOC [7]. However, our dataset reveals a non-monotonic trend at higher SOC across all three temperatures (see Figs. 4–6). For instance, in the 45 °C experiments, cells stored at 70% SOC showed faster degradation than those stored at 100% SOC (Fig. 4). This interesting trend has also been observed in other studies [21,36]. It is attributed to a reversible self-discharge mechanism, caused by oxalates shuttling between the anode and cathode, which diverts lithium away from SEI formation. See [21] for a detailed explanation.

4.2. Differential voltage analysis

DVA is a non-invasive characterisation technique that uses near-equilibrium voltage (V_c) and capacity (Q) data from a slow constant current (dis)charge to generate a differential voltage (DV) plot [39,40]. The DV plot shows dV_c/dQ plotted against Q , and the peaks in the plot represent the phase equilibria of the individual electrodes. Fig. 1 presents the DV plot derived from a $C/20$ discharge data of a fresh LGM-50T cell, with the peaks attributed to the negative and positive electrodes. The process of assigning peaks to individual electrodes is explained in the supplementary material Section S3. In a DV plot, the horizontal distance between peaks corresponding to the same electrode is proportional to the capacity of that electrode [21,41]. Furthermore, the distance between peaks of negative and positive electrodes indicates electrode balancing. Consequently, Q_n and Q_p in Fig. 1 represent the capacities of the negative electrode and positive electrode respectively. Therefore, the LAM of each electrode can be quantified as,

$$\text{LAM}_k = \left(1 - \frac{Q_{k,\text{aged}}}{Q_{k,\text{fresh}}}\right) \times 100\%, \quad (6)$$

where k is either n (negative electrode) or p (positive electrode). Furthermore, Q_{FC} denotes the full cell capacity during the slow discharge.

Figs. 2(a)–2(c) show the ratio of Q_n , Q_p and Q_{FC} between the aged state and the beginning of life. The aged state in these figures corresponds to the final RPTx, conducted on various days between the 483rd and 511th days, for cells stored under different conditions. Each

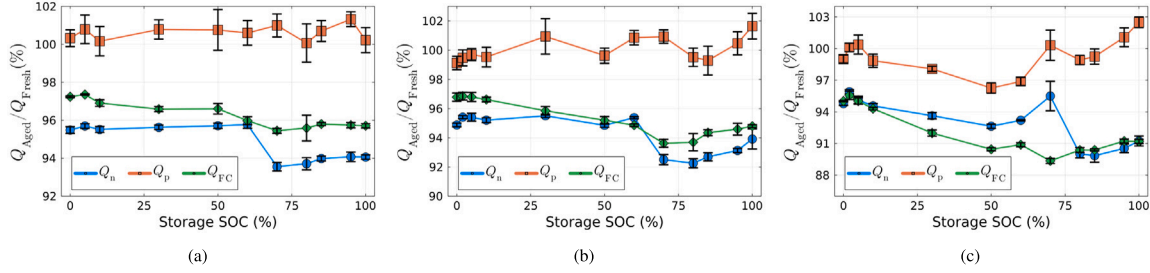


Fig. 2. The ratio of negative electrode capacity (Q_n), positive electrode capacity (Q_p), and full cell capacity (Q_{FC}) between the final RPTx and the first RPTx performed at the beginning of life. The plot illustrates these ratios at different storage SOC and at storage temperatures of (a) 0 °C, (b) 25 °C, and (c) 45 °C.

point in Figs. 2(a)–2(c) is based on DVA performed on slow voltage data from 3 different cells stored under the same condition. The small error bars indicate that these features are minimally affected by measurement noise or errors in peak identification.

For cells stored at 0 °C (Fig. 2(a)) and 25 °C (Fig. 2(b)), the ratio of Q_p stays close to 100%. The slight fluctuations can be attributed to errors during peak identification. The Q_p ratio also remains close to 100% for cells stored at 45 °C, with a slight decrease of up to 3% observed for cells stored at 50% and 60% SOC (Fig. 2(c)). Since, these two are the only cases where positive electrode LAM is observed and the changes are very small, the effect of positive electrode LAM is omitted in this study.

In contrast, the negative electrode capacity (Q_n) shows a significant decrease under all storage conditions considered, with the ratio closely matching the ratio of the total cell capacity (Q_{FC}). This suggests negative electrode LAM is dominant and cannot be disregarded. Zilberman et al. also reported significant negative electrode LAM and minimal positive electrode LAM during calendar ageing of a commercial cell with the same chemistry [36]. The negative electrode LAM can be explained by the SEI side reactions on silicon anode, which generate products that cause pore blockage [42,43]. The pore blockage hinders the access of the electrolyte to the negative electrode particles, resulting in the loss of the negative electrode active material.

SEI side reactions and pore blockage were identified as dominant degradation mechanisms. However, based only on DVA, it is not possible to distinguish between the individual contributions of these two mechanisms to the total drop in cell capacity. Therefore, both of these mechanisms are considered during calendar ageing modelling.

4.3. Physics-based modelling of calendar ageing

Physically-based degradation models consist of an electrochemical model coupled with differential equations that describe the relevant degradation mechanisms. This study uses the single particle model with electrolyte (SPMe) proposed in [4] as the electrochemical model. SPMc assumes that all particles within an electrode behaves identically, allowing them to be represented using a single particle. This approximation is valid for the moderate current rates considered in this study, which do not exceed $C/3$. The SPMc model equations are briefly summarised in Section S1 of the supplementary material for completeness. The reader is referred to [4] for a more detailed understanding of the model. The SPMc model parameters for the LGM50 cells were taken from [34] and are summarised in Table S1.

As discussed in Section 4.2, SEI side reactions and associated pore blockage are identified as the dominant degradation mechanisms. The SEI growth is modelled using the Tafel kinetics limited by the solvent diffusion model proposed in [4]. The model describes the current density of the SEI side reactions (J_{SEI}) at the anode particle surface using Tafel kinetics. Additionally, the reaction rate is also controlled by the solvent diffusion through the SEI layer. Note that, the chosen model can represent two other widely used SEI growth models, the solvent kinetic-limited SEI model [45] and the solvent diffusion-limited

Table 2

Parameters of the physics-based degradation model. The fixed parameters were directly taken from [4] except U_{SEI} which was taken from [44]. The parameters k_{SEI} , D_{SEI} , α_{SEI} , and β are tuned independently at 0 °C, 25 °C and 45 °C using the WMG calendar ageing dataset. See Table 3 for the results.

Symbol	Units	Description	Value
k_{SEI}	$m\ s^{-1}$	Reaction rate of SEI growth	tunable
D_{SEI}	$m^2\ s^{-1}$	Diffusivity of the SEI material	tunable
α_{SEI}	–	Charge transfer coefficient	tunable
β	$m^3\ A^{-1}\ s^{-1}$	Pore clogging rate constant	tunable
$c_{sol,0}$	$mol\ m^{-3}$	SEI material concentration in electrolyte	4541
U_{SEI}	V	Open-circuit potential of SEI reaction	0.4
M_{SEI}	$kg\ mol^{-1}$	Molar weight of SEI material	0.162
ρ_{SEI}	$kg\ m^{-3}$	Density of SEI material	1690
σ_{SEI}	$S\ m^{-1}$	Conductivity of the SEI layer	5×10^{-6}
$L_{SEI,init}$	m	Initial thickness of the SEI layer	5×10^{-9}

SEI model [6,46], through specific parameter choices [47]. The model equations are presented in Eq. (7).

$$J_{SEI} = a_n F k_{SEI} c_{sol} \exp\left(-\frac{\alpha_{SEI} F}{RT} \eta_{SEI}\right),$$

$$\eta_{SEI} = \bar{\eta}_n - U_{SEI},$$

$$c_{sol} = c_{sol,0} - \frac{J_{SEI} L_{SEI}}{a_n F D_{SEI}}.$$

In Eq. (7), c_{sol} represents the solvent concentration at the anode particle surface and L_{SEI} represents the SEI thickness. Further, $\bar{\eta}_n$ represent average anode potential at particle surface calculated using Eq.(S7). The current density of SEI side reactions is added into the boundary condition at the anode surface (Eq.(S1b)) in the SPMc model. Note that, the spatial variation of J_{SEI} was assumed to be negligible. This assumption is valid because, the potential at particle-SEI interface that drives the SEI side reactions become spatially uniform during calendar ageing.

The SEI side reaction causes the SEI layer to grow and anode porosity to decrease, as described by Eqs. (8) and (9) respectively.

$$\frac{dL_{SEI}}{dt} = \frac{1}{2a_n F} \frac{M_{SEI}}{\rho_{SEI}} J_{SEI} \quad (8)$$

$$\frac{de_n}{dt} = -a_n \frac{dL_{SEI}}{dt} \quad (9)$$

Further, to model the anode LAM caused by pore blockage by the SEI products, the linear model proposed in [5] is used. An alternative is the solvent consumption model from [46], which also predicts LAM during SEI growth. However, the solvent consumption model predicts equal LAM in both the anode and cathode, which contradicts our observation that cathode LAM is negligible. The linear pore blockage model relates the change of volume fraction of anode active material ($c_n^a = a_n R_n/3$) to J_{SEI} , as shown in Eq. (10).

$$\frac{dc_n^a}{dt} = -\beta J_{SEI} \quad (10)$$

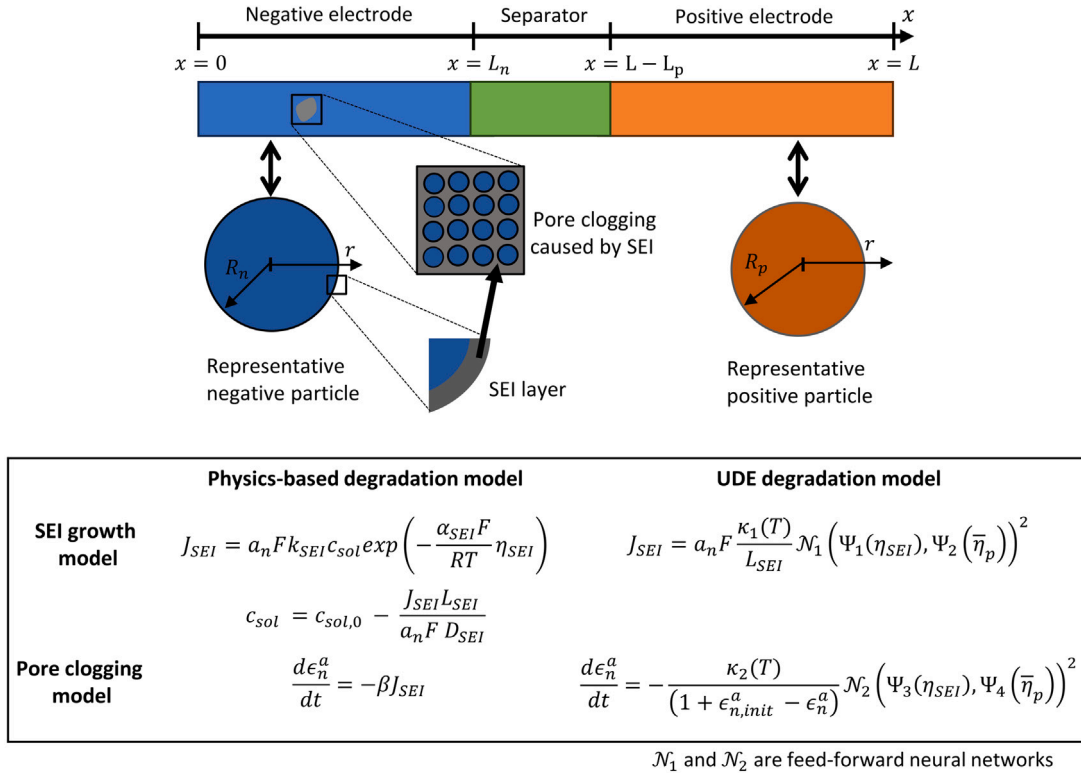


Fig. 3. Schematic of the computational domain for the electrochemical and degradation model. The upper rectangle with 1D Cartesian coordinates $x \in [0, L]$ represents the porous electrode region. Blue and red spheres with radial coordinates $r \in [0, R_k]$ ($k \in \{n, p\}$) denote representative particles in negative and positive electrodes, respectively. The dominant degradation mechanisms, which are the growth of the SEI layer and associated pore clogging in the negative electrode, are also illustrated. Grey patches in the negative electrode, indicate areas where particles become inaccessible for intercalation due to pore clogging. Governing equations describing these degradation mechanisms in physics-based and UDE models are presented below the schematic. See the main text for an explanation of the individual terms in the models.

The physical meanings and values of the parameters in equations Eq. (7) to Eq. (10) is provided in Table 2. Note that, some of these parameters belong to the SPM model, which is given in Table S1. Among the degradation model parameters, k_{SEI} , D_{SEI} , α_{SEI} , and β tunable parameters, while the remaining parameters are material properties with known values taken from literature [4,44]. Furthermore, the negative electrode (anode) LAM predictions can be calculated using:

$$\text{LAM}_n = \left(1 - \frac{\epsilon_n^a}{\epsilon_{n,init}^a}\right) 100\%. \quad (11)$$

Note that, L_{SEI} and ϵ_n^a are the degradation state variables (z in Eq. (1b)) in the physics-based degradation model considered here. Section 5.1 discuss the model parameterisation and validation using the experimental data. The results indicate that the physics-based model correctly explains the square root of time behaviour (self-limiting trend) of the capacity drop qualitatively. However, the values do not match the experimental data accurately for all the storage conditions considered. To address this drawback, we develop a UDE degradation model, as discussed in Section 4.4.

4.4. UDE degradation modelling of calendar ageing

In this Section, we develop a UDE degradation model for calendar ageing. As discussed in Section 2, the electrochemical model remains the same as in the physics-based model, which is the Single Particle Model with Electrolyte (SPMe). The modifications were only made to the degradation equations.

First, we replace the Tafel equation, which predicts the current density for SEI side reactions, with a new equation shown in Eq. (12).

$$J_{SEI} = a_n F \frac{\kappa_1(T)}{L_{SEI}} \mathcal{N}_1\left(\Psi_1(\eta_{SEI}), \Psi_2(\bar{\eta}_p)\right)^2, \quad (12)$$

$$\eta_{SEI} = \bar{\eta}_n - U_{SEI}.$$

Here, $\kappa_1(T)$ is a temperature dependent rate constant, and L_{SEI} is the SEI thickness. We replace the exponential dependency of J_{SEI} on η_{SEI} from Eq. (7) with a neural network $\mathcal{N}_1(\cdot)$. The neural network also takes the spatial average of potential at the cathode particle-electrolyte interface ($\bar{\eta}_p$) as an input, based on findings from [21] suggesting that cathodic side reactions can affect SEI growth rate. Further, Ψ_1 and Ψ_2 are potential feature engineering transformations applied to the inputs η_{SEI} and $\bar{\eta}_p$. Modellers can design these transformations based on physical insights or data trends to enhance the neural network training, resulting in faster error convergence and lower final error values [48]. If no prior insights are available, identity function can be used.

The SEI growth and negative porosity equations remain the same as Eqs. (8) and (9), respectively. Note that, by placing L_{SEI} in the denominator of Eq. (12), we ensure that L_{SEI} has a square root of time dependency. This ensures the self limiting trend of capacity drop as observed in the calendar ageing data. Further, squaring the $\mathcal{N}_1$ output which ensure the SEI growth is irreversible. In practise, this can be implemented by using a square activation function in the output layer.

Further, to improve the anode LAM model, Eq. (10) is replaced with a new differential equation, as shown in Eq. (13).

$$\frac{d\epsilon_n^a}{dt} = -\frac{\kappa_2(T)}{(1 + \epsilon_{n,init}^a - \epsilon_n^a)} \mathcal{N}_2\left(\Psi_3(\eta_{SEI}), \Psi_4(\bar{\eta}_p)\right)^2 \quad (13)$$

Here, κ_2 is a temperature-dependent rate constant, and $\epsilon_{n,init}^a$ is the volume fraction of anode active material at the beginning of life. The DVA analysis shows that anode LAM has a significant effect on the full cell capacity loss. As the full cell capacity loss exhibits a self-limiting trend, we expect the anode LAM to have a self-limiting trend too. To ensure this, we introduce the term $(1 + \epsilon_{n,init}^a - \epsilon_n^a)$ in the denominator. Moreover, based on insights from [23], we expect anode LAM during calendar ageing as a function of SEI growth. Therefore, $\frac{d\epsilon_n^a}{dt}$ is made as

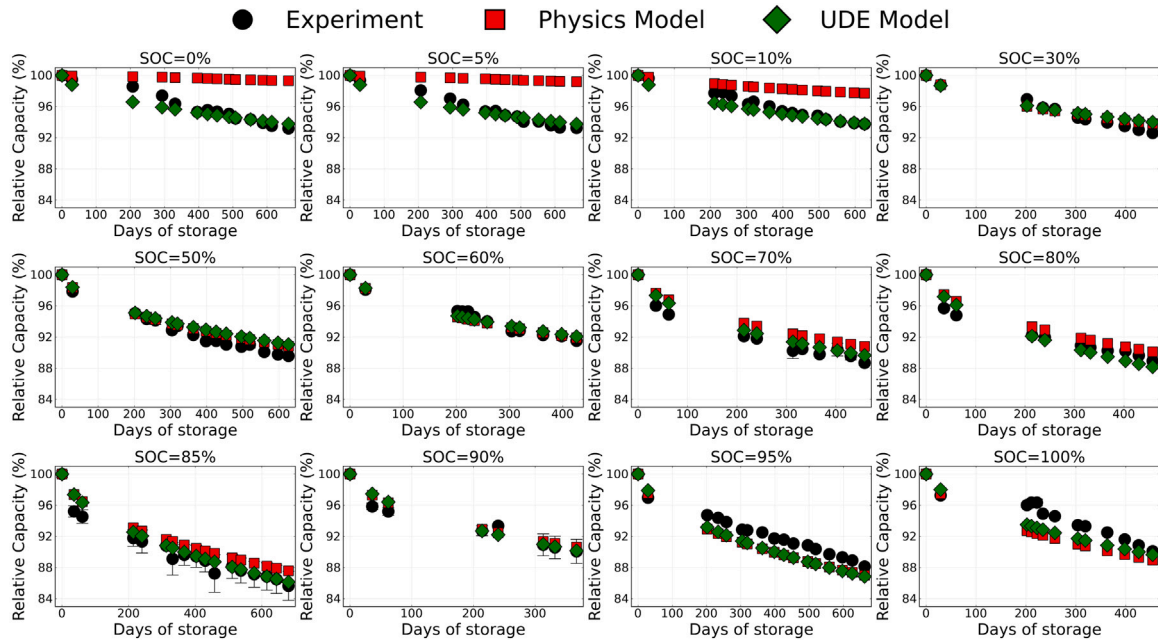


Fig. 4. Comparison of capacity predictions by the physics-based model and UDE model against experimental data for cells stored at 45 °C across different storage SOC values. The storage SOC value is given as title for each sub-figure.

a function of potentials affecting the SEI growth (η_{SEI} and $\bar{\eta}_p$), using the neural network $\mathcal{N}_2(\cdot)$. Ψ_3 and Ψ_4 are suitable feature engineering transformations. Furthermore, squaring the $\mathcal{N}_2$ output ensures ϵ_n^a decreases monotonically with time, which is physically consistent.

Note that, both the degradation rate equations, Eqs. (12) and (13), use rate constants ($\kappa_1(T)$ and $\kappa_2(T)$) to account for the temperature effects. This enables the neural networks $\mathcal{N}_1$ and $\mathcal{N}_2$ to be trained only on data from a single temperature. For other temperatures, only the rate constants need to be adjusted. Section 5.3 validates this design choice, showing that the UDE model trained only on 45 °C data maintains good accuracy at other temperatures through appropriate adjusting of κ_1 and κ_2 .

Fig. 3 shows schematic of the computational domain of the electrochemical model and associated degradation mechanisms. Additionally, it also provides a comparison of governing equations describing these degradation mechanisms in both physics-based and the UDE degradation model.

5. Model parameterisation and validation

5.1. Parameterisation and validation of physics-based model

This Section focuses on the implementation, parameterisation and validation of the physics-based model discussed in Section 4.3. The implementation starts with the spatial discretisation of the SPMe PDEs using the finite volume method (FVM) to derive a system of ODEs. The mesh consists of 10 elements per anode and cathode particle domains, 10 elements in the anode and cathode electrolyte domains, and 6 elements in the separator domain. The resulting ODE system is solved using the Rosenbrock23 solver from the DifferentialEquations.jl library in the Julia programming language [49]. The simulation involved the same conditions used for the ageing experiments, including the RPTs. To simulate the voltage hold phase in the RPTs, the ODE system is solved with constant voltage as an additional algebraic condition.

The next step involves estimating the tunable model parameters: k_{SEI} , D_{SEI} , α_{SEI} , and β , using capacity data from RPTs and LAM data from RPTxs. The loss function $\mathcal{L}_{phy}$ gives more weight to capacity data over LAM due to the higher frequency of RPTs compared to RPTxs. The

Table 3

Optimised values of the tunable parameters of the physics-based degradation model.

Parameter	0 °C	25 °C	45 °C
k_{SEI} (m ² s ⁻¹)	2.928×10^{-16}	1.098×10^{-16}	7.32×10^{-16}
α_{SEI}	0.39	0.5	0.53
D_{SEI} (m ² s ⁻¹)	8.12×10^{-23}	1.856×10^{-22}	1.16×10^{-21}
β (m ³ A ⁻¹ s ⁻¹)	1.385×10^{-9}	1.22×10^{-9}	4.53×10^{-10}

loss function is:

$$\mathcal{L}_{phy} = \sum_{j=1}^M \left[\frac{\omega_2}{M_{j,RPT}} \sum_{i=1}^{M_{j,RPT}} (\Delta Q_{j,i})^2 + \frac{\omega_1}{M_{j,RPTx}} \sum_{i=1}^{M_{j,RPTx}} (\Delta LAM_{j,i})^2 \right] \quad (14)$$

where M is the number of experimental conditions. $M_{j,RPT}$ and $M_{j,RPTx}$ represent the number of RPTs and RPTxs, respectively, for the j th experimental condition. Further, $\Delta Q_{j,i}$ is the error in the $C/3$ capacity predictions for the j th dataset and i th RPT, and $\Delta LAM_{j,i}$ is the error in the j th dataset and i th RPTx. Furthermore, ω_1 and ω_2 are weights, which were chosen to be 0.75 and 0.25, respectively.

Data from cells stored at 30%, 50%, and 80% ($M = 3$) are used for the parameter estimation. To account for the temperature dependency, these parameters are independently estimated using data at 0, 25, and 45 °C. The particle swarm optimiser (PSO) from Metaheuristics.jl was used for performing the optimisation. The PSO parameters used were: population size of 15, cognitive coefficient of 2.0, social coefficient of 2.0, inertia weight of 0.8, and 100 iterations. The optimal parameters are presented in Table 3.

After estimating the model parameters, the physics-based model predictions are validated against the experimental data. Figs. 4–6 shows the comparison between capacity predictions of physics-based model (shown in red) and experimental data (shown in black) for cells stored at temperatures 45 °C, 25 °C, and 0 °C, respectively. In addition, the RMSE of capacity predictions by the physics-based model are shown (in red) in Fig. 7. Similarly, Fig. S7–S9 illustrate the comparison between the loss of active material (LAM) predictions of the physics-based model and the LAM predictions obtained by performing differential voltage analysis (DVA) on the experimental data from cells stored at temperatures 45 °C, 25 °C, and 0 °C, respectively. The corresponding RMSE of the anode LAM predictions are presented in Fig. 8.

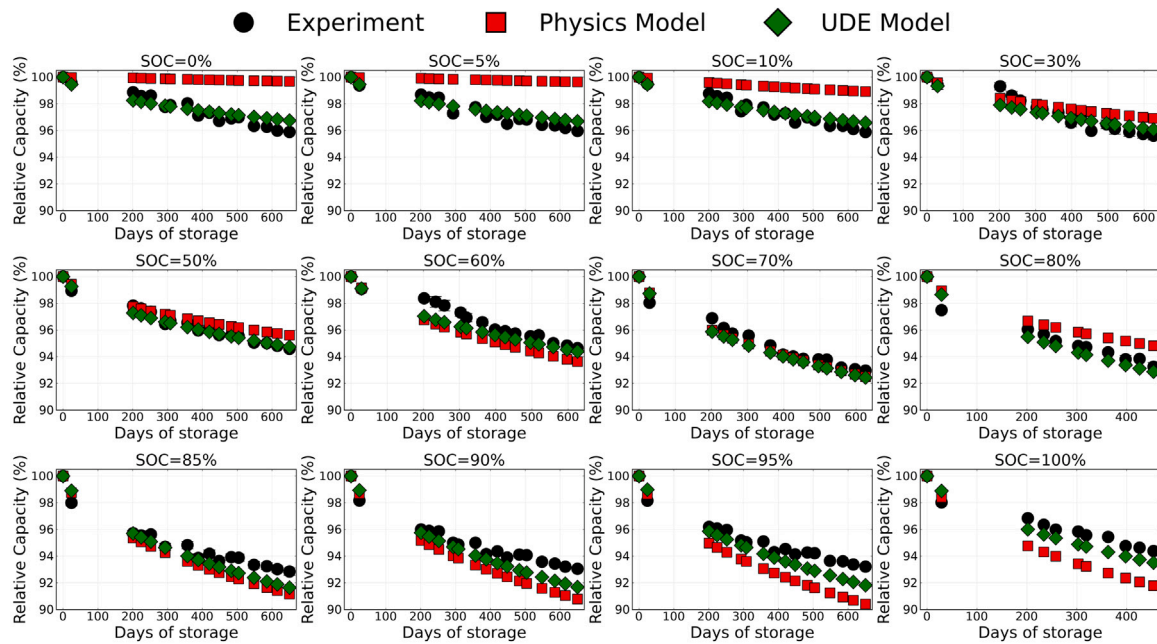


Fig. 5. Comparison of capacity predictions by the physics-based model and UDE model against experimental data for cells stored at 25 °C across different states of charges. The storage SOC value is given as title for each sub-figure.

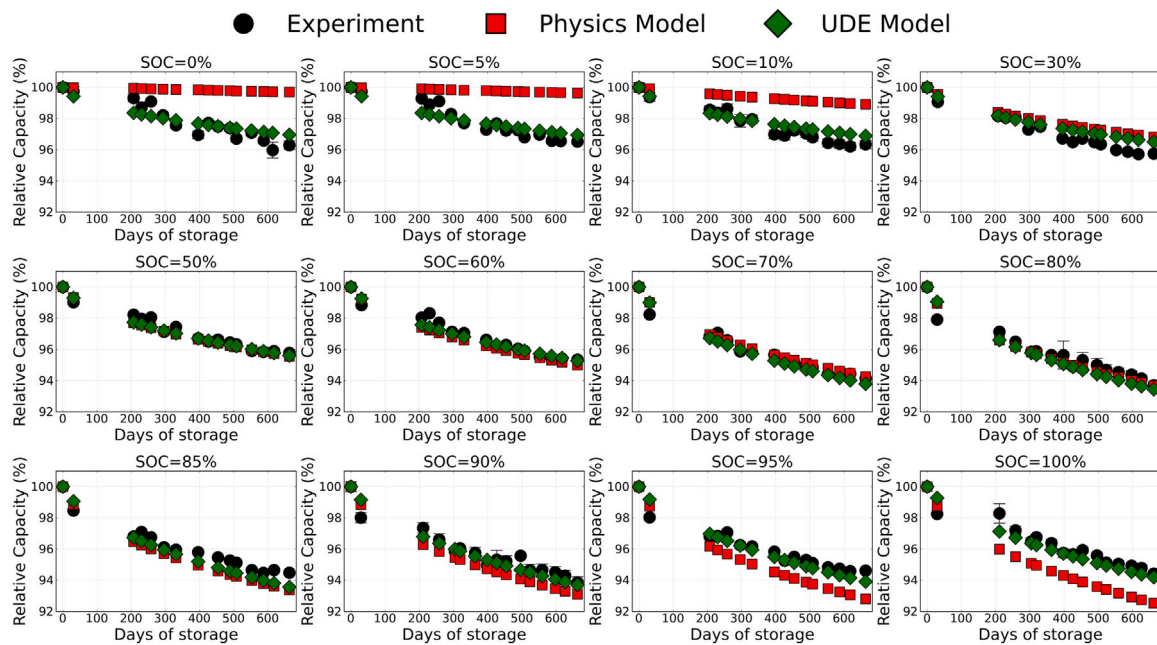


Fig. 6. Comparison of capacity predictions by the physics-based model and UDE model against experimental data for cells stored at 0 °C across different states of charges. The storage SOC value is given as title for each sub-figure.

As shown in Figs. 4–6, the experimental observed capacity trajectory shows a self limiting trend over time, which is a characteristic of SEI growth [4]. The physics-based model predicts this self limiting trend qualitatively, thanks to the solvent diffusion accounted in the SEI growth equations. However, the physics-based model fails to describe the capacity trend quantitatively across all conditions. At all three storage temperatures, the model predictions significantly differ from the experimental data, especially for storage SOC values below 30% and above 80%. Similarly, considering the anode LAM predictions (Fig. S7–S9), the physics-based model offer predictions aligning with experimental data for most SOC values, except at very low SOC values. At storage SOC values below 30%, the model significantly underestimates the anode LAM.

The poor generalisation of quantitative predictions indicate missing physics in the model. Specifically, according to Tafel kinetics, the rate of SEI side reactions has an inverse exponential relationship with SEI potential, which becomes equal to anode equilibrium potential during cell storage. The anode equilibrium potential has a sharp increase at low anode lithiation stoichiometries, particularly for full cell SOC below 30% [34]. Therefore, as per Tafel kinetics, the ageing rate should exhibit substantial reduction when they are stored at SOC values lower than 30% compared to being stored at 30% SOC. However, such a sharp decrease in ageing rate is not observed in the experimental data at all three temperatures considered (see Figs. 4–6).

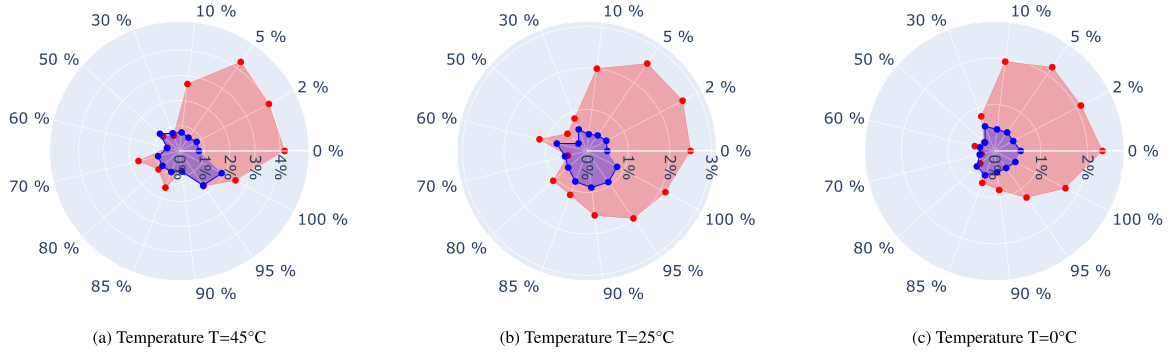


Fig. 7. Radial charts showing RMSE of relative capacity prediction by physics-based model (red) and UDE degradation model (blue). The radial axes show RMSE value in percentage and circumferential axes show storage SOCs. The plots correspond to storage temperatures (a) 45 °C, (b) 25 °C, and (c) 0 °C respectively.

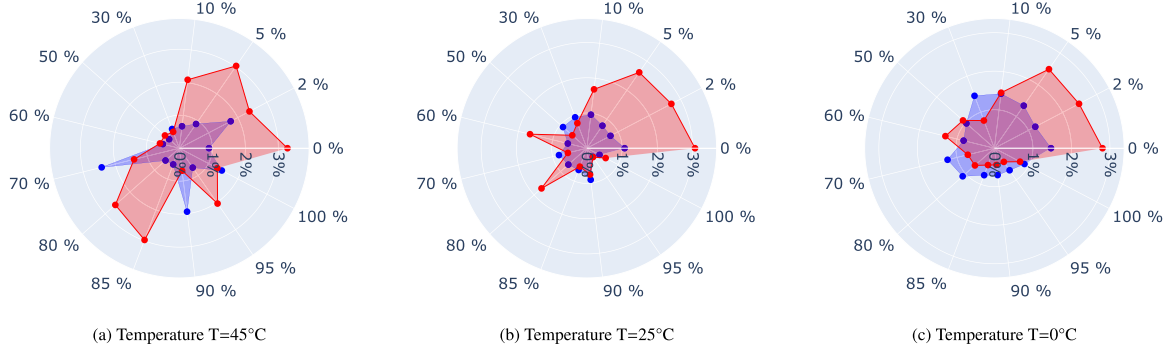


Fig. 8. Radial charts showing RMSE of anode LAM prediction by physics-based model (red) and UDE degradation model (blue). The radial axes show RMSE value in percentage and circumferential axes show storage SOCs. The plots correspond to storage temperatures (a) 45 °C, (b) 25 °C, and (c) 0 °C respectively.

This results in severe under-prediction of capacity loss and LAM at lower storage SOCs, resulting in high RMSE values (see Figs. 7 and 8). Furthermore, the exponential relation also causes the physics-based model to over predict the ageing rate at higher SOCs, especially noticeable for cells stored at 0 °C and 25 °C. It is worth noting that, other continuum SEI models such as the electron tunnelling and interstitial diffusion models also use this exponential dependence [47] and would suffer from similar issues.

To address the issue of the exponential relationship, we attempted to reduce the value of α_{SEI} inside the exponent. Lowering the α_{SEI} value makes the exponential curve more shallow. Although this approach improved accuracy at higher storage SOCs, even decreasing the α_{SEI} to 0.1 could not adequately explain the capacity drop trend observed at lower SOCs. Therefore, in this study, we maintained the values of α_{SEI} obtained from model parameterisation, which were around 0.5. Note that, in majority, if not all, of the previous studies, $\alpha_{SEI} \approx 0.5$ was used. This supports our decision to retain this value.

As detailed above, model validation reveals the inadequacy of the physics-based model to explain the experimental trends adequately. This conclusion is unsurprising as the SEI growth is a highly complex phenomenon involving multiple reactions, whereas the Tafel kinetics model is a simplified representation that considers only a single representative reduction reaction.

5.2. UDE degradation model parameterisation

This Section focuses on the parameterisation of the UDE degradation model proposed in Section 4.4. The parameterisation involves the estimation of the parameters of the neural networks $\mathcal{N}_1$ and $\mathcal{N}_2$, along with the temperature-dependent rate constants κ_1 and κ_2 . To achieve this, we utilise the APTD methodology and the mini-batching approach described in Sections 3.2 and 3.3 respectively. Furthermore, appropriate feature engineering transformations (Ψ_i in Eqs. (12) and

(13)) are also determined in this Section.

Note that, the neural networks $\mathcal{N}_1$ and $\mathcal{N}_2$ do not consider temperature as an input variable. Instead, the influence of temperature on J_{SEI} (Eq. (12)) and derivative of ϵ_n^a (Eq. (13)) is captured through the temperature-dependent rate constants κ_1 and κ_2 , respectively. This formulation allows neural network parameters to be estimated using data from cells stored at a single temperature, which in this study is 45 °C. For other temperatures, such as 0 °C and 25 °C, the same neural network parameters can be retained, and only the rate constants need to be updated. Without losing generality, we can set κ_1 and κ_2 equal to one at 45 °C. For other temperatures, they can be estimated using data from just one or two storage conditions at that temperature.

To estimate neural network parameters, data from eight storage SOCs (5, 30, 50, 60, 70, 80, 85, 95%) at 45 °C were used. To enable faster gradient calculation during the neural network training, the three steps of APTD methodology (see Section 3.2) are followed. The first step is to estimate the degradation states, L_{SEI} and ϵ_n^a , as a function of time for the ageing datasets considered. To achieve this, a nuanced strategy different from the suggested approaches in Section 3.2 is used, which is detailed below.

As discussed in Section 5.1, the physics-based model captures the qualitative trend of the cell capacity and anode LAM as observed in the experimental data. But, it fails to quantitatively explain the trends observed at different storage conditions using a single parameter set. However, as the qualitative behaviour matches, for any individual storage condition, it was possible to estimate a set of tunable parameters that accurately describe both the capacity drop and LAM trends quantitatively. Note that, when the observable capacity drop and LAM trends matches, it also means the model correctly explains the evolution of the degradation states for that particular ageing condition.

This means, the physics-based model can be independently tuned for each storage datasets included for neural network training to estimate the degradation state evolution for that specific ageing condition ($\hat{z}$ in

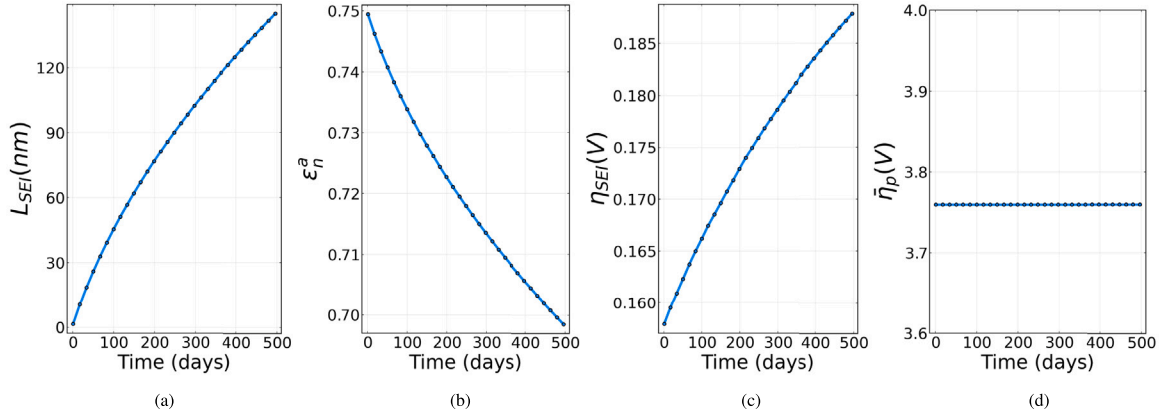


Fig. 9. Augmented dataset corresponding to calendar ageing at 30% storage SOC and 45 °C temperature. (a) SEI length, (b) Anode active material volume fraction, (c) SEI overpotential, (d) cathode potential.

Section 3.2). Moreover, this approach can also provide estimates of the electrochemical states ($\hat{x}$ in Section 3.2) needed for the construction of the augmented dataset, which is the second step in APTD. However, as noted in Section 3.2, we do not need to save all the electrochemical states in the augmented dataset, but rather a lower dimensional mapping of it. In this study, it is just two quantities, namely η_{SEI} and $\bar{\eta}_p$, as can be seen in the UDE degradation equations Eqs. (12) and (13). Based on this observation, we can create the augmented dataset for an ageing dataset indexed by j as $D_j = \{t, L_{SEI}(t), \epsilon_a^a(t), \eta_{SEI}(t), \bar{\eta}_p(t)\}$.

Following the above mentioned method, the parameters k_{SEI} , D_{SEI} , α_{SEI} , and β are tuned independently for the eight ageing conditions selected for neural network training. The parameters were first manually optimised, followed by fine tuning using the PSO algorithm. After obtaining the optimal parameters, the augmented datasets for each condition were constructed. All the time series components in the augmented dataset showed small variations over time, except for RPTs, where the potentials $\eta_{SEI}(t)$ and $\bar{\eta}_p(t)$ showed significant variation. However, the RPT based variations were removed before adding them to the augmented dataset. This was based on the assumption that degradation primarily takes place during storage rather than during RPTs. After removing the RPT data, only 30 data-points were required to capture the time evolution of the augmented dataset components. Fig. 9 shows an example of the augmented dataset corresponding to the storage at 30% SOC and 45 °C. Following the construction of the augmented dataset, the next step in APTD methodology is the problem reformulation. A new time variable $\tau = t/\gamma$ is introduced, where one unit of τ equals one day ($\gamma = 86400$). Further, the ageing duration is divided into disjoint segments. The segments were chosen to be one day long, based on the observation that the degradation states, L_{SEI} and ϵ_a^a , remain nearly constant within a day (see Fig. 9 for example). Thus, the function mapping τ to its corresponding time segment in t can be defined as $\sigma(\tau) = [\gamma\tau, \gamma(\tau+1)]$.² As shown in Eq. (3), the SEI growth model in the UDE degradation model (Eqs. (12) and (8)) can be reformulated as

$$\frac{dL_{SEI}}{d\tau} = \left(\frac{1}{2} \frac{M_{SEI}}{\rho_{SEI}} \right) \int_{\gamma\tau}^{\gamma(\tau+1)} \frac{\kappa_1(T)}{L_{SEI}} \mathcal{N}_1(\Psi_1(\eta_{SEI}), \Psi_2(\bar{\eta}_p))^2 dt \quad (15)$$

Further simplifications can be made to Eq. (15). The integration in the right hand side is performed over a day, during which the variables inside the integrand (η_{SEI} , $\bar{\eta}_p$, L_{SEI}) show minimal variation during cell storage. Therefore, Eq. (15) can be rewritten as

$$\frac{dL_{SEI}}{d\tau} = \left(\frac{\gamma}{2} \frac{M_{SEI}}{\rho_{SEI}} \right) \frac{\kappa_1(T)}{L_{SEI}} \mathcal{N}_1(\Psi_1(\eta_{SEI}), \Psi_2(\bar{\eta}_p))^2. \quad (16)$$

² $\lfloor x \rfloor$ denotes the floor function (largest integer $\leq x$) and $\lceil x \rceil$ denotes the ceiling function (smallest integer $\geq x$).

The reformulated Eq. (16) enable cost efficient training of the $\mathcal{N}_1$ parameters. Note that, for the ageing datasets considered for $\mathcal{N}_1$ training, the functions $\eta_{SEI}(t)$ and $\bar{\eta}_p(t)$ are precomputed and stored in the augmented dataset. Therefore, while solving Eq. (16) for the adjoint calculations, these functions can be directly accessed from the augmented dataset, instead of solving a coupled electrochemical model in parallel. This decoupling helps the solver to take time steps in the degradation time scale while solving Eq. (16), thereby addressing the parameterisation challenge detailed in Section 3.

The loss function for $\mathcal{N}_1$ training is formulated as the mean squared error (MSE) between solution of Eq. (16) and L_{SEI} values from the augmented dataset. During its optimisation, a mini-batching approach is followed, where the loss function considers only one of the eight datasets per iteration to calculate the gradient. Mini-batching reduces the computational and memory costs while also helping the optimiser to avoid local minima during training.

The quadrature adjoint technique [13] from SciMLSensitivity.jl was used for the gradient calculation of the loss function. Additionally, the Jacobian computations required for the adjoint calculations were carried out using reverse-mode AD technique implemented in Zygote.jl library [50]. Furthermore, the required ODE solving was performed using TSit5 solver from DifferentialEquations.jl library with 10^{-6} solver tolerance [49].

The architecture of neural network $\mathcal{N}_1$ consisted of an input layer with two neurons using a linear activation function, followed by two hidden layers with 10 neurons each and tanh activation functions, and an output layer with a single neuron using a square activation function. The ADAM optimiser from Optim.jl with a learning rate of 1×10^{-3} and 200 epochs (1600 iterations) was used for $\mathcal{N}_1$ training [51]. The optimisation was performed on a computer with Intel Xeon Gold 6148 CPU and 96 GB of RAM. With the APTD problem reformulation and mini-batching, the total optimisation of $\mathcal{N}_1$ parameters took about 30 min.

To determine suitable feature engineering transformations, (Ψ_1 and Ψ_2) in Eq. (16), the following approach was used. First, these transformations were chosen based on domain knowledge or insights from data. Following this, a comparison is made between results from $\mathcal{N}_1$ training using both the chosen functions and the identity functions. If the chosen transformations offered lower optimal error, they were retained, otherwise identity functions were used instead.

In the case of Ψ_1 , we chose an identity function ($\Psi_1(\eta_{SEI}) = \eta_{SEI}$) to reflect our uninformed prior insights. Further, regarding Ψ_2 , as per [21], the influence of cathode potential on SEI growth is due to side reactions at cathode occurring at cathodic potentials above 4.1 V [44]. Therefore, Ψ_2 was chosen as the rate of cathode side reactions modelled using tafel kinetics as $\Psi_2 = \exp\left(\frac{0.5F}{RT}(\bar{\eta}_p - 4.1)\right)$. This choice of Ψ_2 resulted in faster error convergence and lower optimal error, hence it

was retained.

The strategies used for $\mathcal{N}_1$ training were also used for $\mathcal{N}_2$ training as well. Following the same steps, the ODE describing evolution of ϵ_n^a can be reformulated as:

$$\frac{d\epsilon_n^a}{d\tau} = -\gamma \frac{\kappa_2(T)}{(1 + \epsilon_{n,\text{init}}^a - \epsilon_n^a)} \mathcal{N}_2(\Psi_3(\eta_{\text{SEI}}), \Psi_4(\bar{\eta}_p))^2 \quad (17)$$

which can be used for $\mathcal{N}_2$ parameter estimation. The numerical settings, mini-batching approach and neural network architecture remain the same as that of $\mathcal{N}_1$ training, and not repeated for brevity. The transformation Ψ_4 was chosen same as Ψ_2 as cathodic side reaction affects SEI growth and SEI and LAM are linked. However, this choice did not offer lower optimal error than identity function, hence $\Psi_4(\bar{\eta}_p) = \bar{\eta}_p$ was chosen instead.

For Ψ_3 , insights from training data were used. The time series data for ϵ_n^a in the augmented dataset showed that $\epsilon_n^a(t)$ at 5% and 30% storage SOC were very similar. In contrast, it varied considerably between storage SOC at higher SOC. However, note that the η_{SEI} becomes equal to the anode equilibrium potential (U_n in Eq.(S10b)) during cell storage. U_n shows a drastic change with SOC at low SOC and only slight change at higher SOC (see Fig.S2a). This indicates that the change in ϵ_n^a is much more sensitive to η_{SEI} at higher SOC than at lower SOC. To capture this trend, following equation is used for Ψ_3 ,

$$\Psi_3(\bar{\eta}_{\text{SEI}}) = \exp(p_1 \times \text{LeakyReLU}(p_2 - U_{\text{SEI}} - \eta_{\text{SEI}}, p_3))$$

$$\text{LeakyReLU}(y, a) = \begin{cases} y & \text{if } y \geq 0 \\ a \cdot y & \text{if } y < 0 \end{cases} \quad (18)$$

where p_1, p_2 and p_3 are hyper-parameters. The LeakyRelu function ensures that for anode potential above p_2 , the sensitivity can be tuned using the parameter p_3 . The value of p_2 was chosen to be 0.16 which is equal to the anode equilibrium potential at 30% SOC and the slope parameter is set to 0.05. Further, the exponential function ensures higher sensitivity of the derivative of ϵ_n^a to SEI potential at higher SOC. The hyper parameter p_1 was selected to be 10 based on grid search. The function Eq. (18) offered lower optimal error than identity function, hence was retained.

After training both the neural network using data from cells stored at 45 °C, the next step was to estimate the temperature dependent rate constants at 25 °C and 0 °C. To achieve this, we used capacity and anode LAM data from cells stored at 10% and 80% SOC at these two temperatures. Further, the rate constants were independently parameterised at these two temperatures using a cost function same as Eq. (14). The optimised values were $\kappa_1 = 0.32$ and $\kappa_2 = 0.46$ for 25 °C and $\kappa_1 = 0.31$ and $\kappa_2 = 0.245$ for 0 °C.

5.3. UDE degradation model validation

This Section validates the UDE degradation model against experimental data and compares its prediction accuracy with the physics-based model. Figs. 4–6 show a comparison of cell capacity predictions by both models for cells stored at 45 °C, 25 °C, and 0 °C, respectively. Similarly, Fig. S7-S9 compare the anode LAM predictions by both the models against experimental data for the same storage temperatures. Additionally, a comparison of the RMSE for model predictions on cell capacity and anode LAM is provided in Fig. 7 and Fig. 8, respectively.

First, the results from the 45 °C storage data, where the neural networks were trained, are analysed. Focusing on the capacity predictions (Fig. 4 and Fig. 7(a)), the UDE model offered stronger alignment with the experimental observations compared to the physics-based model for majority of the storage SOC. Exceptions are 30% and 50% storage SOC, where the physics-based model offer marginally lower RMSE in predictions (see Fig. 7). An important result is that, the UDE model effectively addresses the severe under prediction of ageing rates observed in physics-based models for storage SOC below 30% (Fig. 4). Overall, for cells stored at 45 °C, the RMSE of the capacity predictions by the UDE model was less than 1.89%. Further, the average RMSE across all

storage SOC were 0.94%, which is a 53.4% reduction compared to the physics-based model.

Next, the results from the anode LAM predictions made for cells stored at 45 °C are analysed (Fig.S1, Fig. 8). The anode LAM predictions by the UDE model closely matched the LAM values calculated from the experimental data using DVA. Similar to the capacity predictions, the UDE model resolves the severe under prediction of anode LAM by the physics-based model for storage SOC below 30%. Further, across all the storage SOC except at 70% and 90% 45 °C, the UDE model offered significantly lower RMSE compared to the physics-based model.

At 70% storage SOC, the DVA LAM estimate for the fourth RPTx (see Fig.S7) is lower than the third, which is physically unlikely. This could be due to errors in peak finding during DVA, as evidenced by the large variance in LAM estimate for the fourth RPTx. Therefore, the high RMSE at 70% storage SOC can be attributed to error in DVA on fourth RPTx. The LAM predictions by the UDE model until the third RPTx matches the DVA estimates better than the physics-based model, highlighting the reliability of UDE model predictions. For cells stored at 90%, the cells showed knee effect and reached end-of-life before the fourth RPTx resulting in only three data points in Fig. S3. These cells behaved differently from the rest of the batch, and can be treated as an outlier.

Overall, for cells stored at 45 °C, ignoring 70% and 90% storage SOC, the RMSE of the anode LAM predictions by the UDE model was less than 1.75%. Further, the average RMSE across all storage SOC were 0.81%, which is a 57.9% reduction compared to the physics-based model.

As discussed above, the UDE model outperformed the physics-based model for calendar ageing at 45 °C, where the neural networks were trained. However, the true test for the reliability of the UDE model is to validate its ability to generalise to scenarios outside the training conditions. To this end, we analyse the UDE model predictions for calendar ageing at 25 °C and 0 °C. For these temperatures, the neural network parameters from 45 °C training were retained, and only the temperature dependent rate constants (κ_1 and κ_2) were adjusted using data from 10% and 80% storage SOC at these temperatures.

The capacity predictions by the UDE model at 25 °C (Fig. 5) and 0 °C (Fig. 6) closely match the experimental data and are considerably more accurate than the physics-based model (see Fig. 7(b) and Fig. 7(c)). At 25 °C, the UDE model achieved an average RMSE of 0.61% and a maximum RMSE of 0.91%, which shows a 61.2% and 65% decrement respectively compared to the physics-based model. Similarly, at 0 °C, the UDE model achieved an average RMSE of 0.45% and a maximum RMSE of 0.56%, representing improvements of 61.2% and 75.6%, respectively, compared to the physics-based model. These findings successfully demonstrate the ability of the UDE degradation modelling approach to learn the underlying physics, which enable it to generalise well to out-of-training-sample scenarios.

Similar to capacity predictions, the anode LAM predictions by the UDE model closely align with the experimental data obtained from calendar ageing at 25 °C (Fig.S8) and 0 °C (Fig.S9). In comparison to the physics-based model, the LAM predictions by the UDE model generally shows lower RMSE, with a few exceptions where the physics-based model shows marginal improvements (see Fig. 8(b) and Fig. 8(c)). The maximum RMSE of the UDE model was 1.07% at 25 °C and 1.46% at 0 °C, which represent a reduction of 44.5% and 11.3% respectively compared to the physics-based model. The high accuracy in LAM predictions also demonstrate the ability of the UDE model to generalise to out-of-training-sample scenarios.

The above analysis indicate that, the UDE model demonstrates strong predictive capabilities across all the operating conditions, including scenarios outside the training data. More importantly, it outperforms the typically used physics-based model across most of the storage conditions. The cases where physics-based model offer a lower RMSE, the decrement is marginal. Additionally, the UDE model resolves the severe under prediction of capacity loss and anode LAM by the

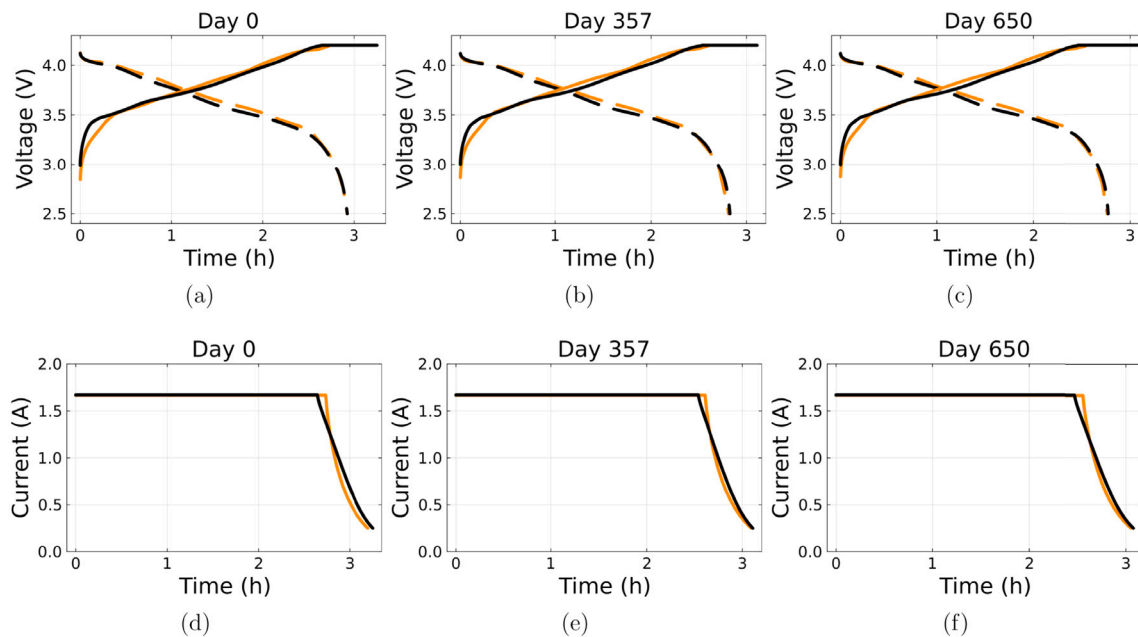


Fig. 10. Voltage predictions by the UDE model (orange) versus experimental data from RPTs (black) for storage durations of (a) 0 days, (b) 357 days, and (c) 650 days for cells stored at 50% SOC and 25 °C. Dashed lines represent C/3 constant current (CC) discharge, while solid lines represent C/3 constant current charging, followed by a voltage hold at 4.2 V until the current decreases to C/20. Subfigures (d), (e), and (f) show the comparison of the UDE model predictions (orange) and experimental data (black) for the corresponding CC-CV charging current profiles.

physics-based model at low storage SOC.

In addition to degradation indicators, the UDE model also predicts evolution of the electrochemical dynamics as the cell ages. To demonstrate this, Fig. 10 presents validation of the voltage and current evolution from RPTs conducted at the beginning of life, on the 357th day, and on the 650th day for a cell stored at 50% SOC and 25 °C. The voltage predictions of the UDE model show good agreement with the experimental results, with an average RMSE of 41.55 mV and a maximum RMSE of 46.8 mV. Furthermore, the UDE model effectively reproduces the current evolution during the constant voltage hold phase, implemented using voltage hold as an additional algebraic constraint.

6. Discussion

In this section, we compare UDE calendar ageing modelling with existing approaches: machine learning (ML) [52], empirical [53], and physics-based models. A comparison between these approaches indicate that ML models offered the highest degree of accuracy in terms of predictions [20]. ML models use flexible function approximators to find correlations between stress factors (SOC, temperature, storage time) with capacity and resistance, directly from the data.

Despite their accuracy, ML model predictions are constrained to the training data, lack physical insights and generalisability. For instance, ML models cannot predict changes in the capacity trajectory if a different experimental protocol was used, such as different C-rate for the capacity measurement or conducting experiments under constant voltage condition instead of open circuit used for model training [54]. More importantly, ML models cannot predict how the electrochemical model behaviour evolves as the cell ages. They cannot capture changes in the SOC-OCV relationship or estimate how much energy can be extracted under specific usage conditions in an aged state.

Unlike ML models, physics-based models describe the physical mechanisms causing ageing. By performing appropriate simulations, the modeller can answer the above-mentioned questions. More importantly, predictions of physics-based models are not limited to just capacity; they can predict the evolution of all electrochemical and

degradation state variables, including those that are not directly measurable. However, as shown in this article, the reliability of physics-based model predictions depends on accurately capturing the relevant physics. If the physics are incorrect, no amount of parameterisation data can improve prediction accuracy.

In the UDE model, the ability of ML models to approximate functions is used to find the relationship between state variables and their derivatives of the physics-based model. This approach ensures that the UDE model achieves high predictive accuracy of ML models while also offering physical insights that the physics-based model can offer. In addition, since the ML model is placed within the constraints of the physics-based model, the function it learns generalises well to scenarios beyond the training data.

A UDE approach also inherits some drawbacks of both methods, particularly regarding the data requirements for model parameterisation. Like physics-based models, the UDE model also require specific electrochemical parameters that often rely on expensive experiments. In addition, it will also require extensive ageing datasets, as an ML model does, to approximate the incomplete model structure for degradation dynamics.

However, note that, the requirement for a large amount of ageing data is not exclusive to UDE models. When the knowledge of the underlying physics is incomplete, traditional scientific approaches will also require extensive experimental data to learn the missing physics and incorporate it into the physics-based model structure. A UDE framework is particularly valuable in this context, as it applies machine learning to automate/assist this learning process. A UDE can also facilitate data-driven model discovery. Algorithms such as SINDy [55] enable the representation of the function (learnt by the neural networks) in a sparse parsimonious manner [13]. This form can be interpreted by experts in the domain to extract meaningful physical insights.

Another important point to note is that, as the UDE approach learns missing physics from experimental data, there is a risk that the model could learn physically implausible trends or noise present in the dataset. To mitigate this risk, the modeller must take preventive measures during data collection, model development, parameterisation, and model verification.

Data Collection: Multiple repeats of experiments should be performed to verify the validity of an observed trend. This is a standard practise in ageing studies, including the presented calendar ageing dataset, where at least 3 cells are tested under a specific ageing condition. In addition, data cleaning should be done to remove potential outliers.

Model development: Unlike purely data-driven models, most UDE components are physically interpretable, allowing the modeller to impose hard physical constraints to ensure that the neural networks in the UDE do not learn physically implausible trends. General examples of such constraints include symmetry, monotonicity, and smoothness [56].

In this study, we used several such constraints. For example, placing L_{SEI} in the denominator of Eq. (12) constrained SEI growth to be self-limiting. Furthermore, we squared the output of $\mathcal{N}_1$ in Eq. (12) to constrain the growth of SEI to be irreversible. These constraints also impose smoothness on the predictions, thereby reducing the influence of noise in capacity measurements on the model predictions.

Parameterisation: During model training, physics-based regularisation can be added to the loss function. This penalise the neural networks from learning physically infeasible solutions [56]. Furthermore, machine learning strategies such as early stopping and dropout can also be employed to prevent overfitting [57].

Model verification and validation: Validation with out-of-sample data should be carried out to confirm its generalisability. Generalisability in validation ensures that the model did not just memorise the training data (overfitting), but has learnt the underlying dynamics. In this study, we only used 8 out of 39 datasets for model training, and the model generalised very well to validation conditions, which is good indication that it did not overfit.

7. Conclusion

This article introduces the Universal Differential Equation (UDE) modelling approach for lithium-ion battery degradation modelling as an alternative to commonly used physics-based and pure-data-driven degradation modelling approaches. Our approach involves incorporating neural networks into the structure of continuum physics-based degradation models. This allows the neural network, when trained with experimental data, to approximate functions representing degradation mechanisms that are incompletely modelled in the physics-based degradation model.

Therefore, the UDE approach is particularly useful in scenarios where sufficient ageing data is available, but physics-based models, even with extensive parameterisation, fail to quantitatively fit the experimental data or replicate qualitative trends in it, due to inherent limitations in their model structure. The UDE framework addresses these shortcomings by learning optimal constitutive relations for poorly modelled degradation mechanisms within the physics-based degradation model directly from data, thereby enhancing its model structure. As a result, the UDE degradation model combines the function approximation capabilities of machine learning with the interpretability of physics-based models.

However, for the battery community to adopt the UDE approach, this article emphasises that the high cost associated with the parameterisation of the UDE model needs to be addressed. To this end, we combined the newly developed APTD method and mini-batching, which enables faster and memory-efficient parameterisation.

Furthermore, by using calendar ageing as an example, this article demonstrates how to formulate UDE degradation models and then parameterise the resultant model using the newly proposed method. After parameterisation, the UDE calendar ageing model was validated using an extensive in-house calendar ageing dataset that spans 39 storage conditions (13 storage SOC's at 0 °C, 25 °C and 45 °C), with an average storage period of 686 days and three cells per condition.

Model validation showcases the ability of the UDE model to capture the degradation behaviour across all operating conditions. The average RMSE across all storage SOC's for capacity prediction, by the UDE model, were 0.94%, 0.61%, and 0.45% at 45 °C, 25 °C, and 0 °C, respectively. In comparison, the widely used physics-based model had higher RMSE of 2.02%, 1.57% and 1.16% respectively. The UDE model significantly outperformed the physics-based model in terms of accuracy of capacity predictions as well as anode LAM predictions. Note that, the neural networks in the UDE model were only trained using the 45 °C dataset. The high accuracy across the other two temperatures demonstrates the ability of the neural networks to learn the underlying physics, which enable generalisation to out-of-training-sample scenarios.

CRediT authorship contribution statement

Jishnu Ayyangatu Kuzhiyil: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Theodoros Damoulas:** Writing – original draft, Supervision, Resources, Methodology. **Ferran Brosa Planella:** Writing – review & editing, Supervision. **W. Dhammika Widanage:** Writing – original draft, Supervision, Software, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the Doctoral College at the University of Warwick for funding the work through the provision of the Chancellor's International Scholarship. FBP was supported by the Faraday Institution, UK [EP/S003053/1 grant number and FIRG059]. For the purpose of open access, the authors have applied a Creative Commons Attribution (CC BY) licence to any Author Accepted Manuscript version arising from this submission.

Appendix A. Summary of adjoint calculations for a system of ordinary differential equations

Consider the ODE equation, with initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t; \boldsymbol{\theta}) \quad t \in [t_0, T] \quad (\text{A.1})$$

and an L_2 norm based discrete cost function defined as:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\theta}) = \sum_{i=1}^N \|\mathbf{x}(t_i, \boldsymbol{\theta}) - \mathbf{x}_{\text{obs}}(t_i)\|^2 \quad (\text{A.2})$$

The adjoint method allows the calculation of the gradient of the loss function with respect to the parameters $\frac{d\mathcal{L}}{d\boldsymbol{\theta}}$ by following the 3 steps given below:

1. Solve the ODE in Eq. (A.1) in the forward time direction (from $t = t_0$ to $t = T$) with the initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$ to obtain $\mathbf{x}(t)$.
2. Solve the adjoint ODE in Eq. (A.3) in the reverse time direction (from $t = T$ to $t = t_0$) with the initial condition $\boldsymbol{\lambda}(T) = \mathbf{0}$ to obtain $\boldsymbol{\lambda}(t)$.

$$\frac{d\boldsymbol{\lambda}}{dt} = - \left(\frac{d\mathbf{f}}{d\mathbf{x}} \right)^T \boldsymbol{\lambda} + 2(\mathbf{x}(t, \boldsymbol{\theta}) - \mathbf{x}_{\text{obs}}(t_i))\delta(t - t_i) \quad (\text{A.3})$$

where δ is the Dirac delta function.

3. Calculate the gradient via:

$$\frac{d\mathcal{L}}{d\boldsymbol{\theta}} = \int_{t_0}^T \left(\boldsymbol{\lambda}^T \frac{d\mathbf{f}}{d\boldsymbol{\theta}} \right) dt \quad (\text{A.4})$$

For the derivation and details about the numerical implementation of the above equations, readers are referred to [13,27].

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.apenergy.2024.125221>.

Data availability

The Julia code for the models used in this article is available on the GitHub repository <https://github.com/JishnuKuzhiyil/Calendar-Ageing-Modelling-using-UDE---Julia-Code.git>, and the experimental calendar ageing dataset is provided in the Zenodo repository <https://zenodo.org/records/14577286>.

References

- [1] Brosa Planella F, Ai W, Boyce AM, Ghosh A, Korotkin I, Sahu S, et al. A continuum of physics-based lithium-ion battery models reviewed. *Prog Energy* 2022;4(4).
- [2] Ali HAA, Raijmakers LH, Chayambuka K, Danilov DL, Notten PH, Eichel RA. A comparison between physics-based Li-ion battery models. *Electrochim Acta* 2024;493.
- [3] Sulzer V, Marquis SG, Timms R, Robinson M, Chapman SJ. Python battery mathematical modelling (PyBaMM).
- [4] Brosa Planella F, Widanage WD. A Single Particle model with electrolyte and side reactions for degradation of lithium-ion batteries. *Appl Math Model* 2023;121:586–610.
- [5] Reniers JM, Mulder G, Howey DA. Review and Performance Comparison of Mechanical-Chemical Degradation Models for Lithium-Ion Batteries. *J Electrochem Soc* 2019;166(14):A3189–200.
- [6] O'Kane SE, Ai W, Madabattula G, Alonso-Alvarez D, Timms R, Sulzer V, et al. Lithium-ion battery degradation: how to model it. *Phys Chem Chem Phys* 2022;24(13):7909–22.
- [7] Edge JS, O'Kane S, Prosser R, Kirkaldy ND, Patel AN, Hales A, et al. Lithium ion battery degradation: what you need to know. *Phys Chem Chem Phys* 2021;23(14):8200–21.
- [8] Han X, Lu L, Zheng Y, Feng X, Li Z, Li J, et al. A review on the key issues of the lithium ion battery degradation among the whole life cycle. *ETransportation* 2019;1.
- [9] Sulzer V, Mohtat P, Aitio A, Lee S, Yeh YT, Steinbacher F, et al. The challenge and opportunity of battery lifetime prediction from field data. *Joule* 2021;5(8):1934–55.
- [10] Cuomo S, Di Cola VS, Giampaolo F, Rozza G, Raissi M, Piccialli F. Scientific Machine Learning Through Physics-Informed Neural Networks: Where we are and What's Next. *J Sci Comput* 2022;92.
- [11] Huang L, Vranceanu D, Wang Y, Kulathunga N, Ranasinghe N. Discovering Nonlinear Dynamics Through Scientific Machine Learning. In: *Lecture notes in networks and systems*, vol. 294, Springer Science and Business Media Deutschland GmbH; 2022, p. 261–79.
- [12] Mistry A, Franco AA, Cooper SJ, Roberts SA, Viswanathan V. How Machine Learning Will Revolutionize Electrochemical Sciences. *ACS Energy Lett* 2021;6(4):1422–31.
- [13] Rackauckas C, Ma Y, Martensen J, Warner K, Zubov K, Supekar R, et al. Universal Differential Equations for Scientific Machine Learning. 2020.
- [14] Kuzhiyil JA, Damoulas T, Widanage WD. Neural equivalent circuit models: Universal differential equations for battery modelling. *Appl Energy* 2024;371.
- [15] Bills A, Sripad S, Fredericks WL, Guttenberg M, Charles D, Frank E, et al. Universal Battery Performance and Degradation Model for Electric Aircraft. 2020.
- [16] Ramasubramanian S, Schomburg F, Röder F. Modelling of solid electrolyte interphase growth using neural ordinary differential equations. *Electrochim Acta* 2024;473:143479.
- [17] Sulzer V, Mohtat P, Pannala S, Siegel JB, Stefanopoulou AG. Accelerated Battery Lifetime Simulations Using Adaptive Inter-Cycle Extrapolation Algorithm. *J Electrochem Soc* 2021;168(12):120531.
- [18] Stapor P, Schmiester L, Wierling C, Merkt S, Pathirana D, Lange BM, et al. Mini-batch optimization enables training of ODE models on large-scale datasets. *Nature Commun* 2022;13(1).
- [19] Keil P, Schuster SF, Wilhelm J, Travi J, Hauser A, Karl RC, et al. Calendar Aging of Lithium-Ion Batteries. *J Electrochem Soc* 2016;163(9):A1872–80.
- [20] Liu K, Ashwin TR, Hu X, Lucu M, Widanage WD. An evaluation study of different modelling techniques for calendar ageing prediction of lithium-ion batteries. *Renew Sustain Energy Rev* 2020;131.
- [21] Zülke A, Li Y, Keil P, Burrell R, Belaisch S, Nagarathinam M, et al. High-Energy Nickel-Cobalt-Aluminium Oxide (NCA) Cells on Idle: Anode- versus Cathode-Driven Side Reactions. *Batteries Supercaps* 2021;4(6):934–47.
- [22] Khaleghi Rahimian S, Forouzan MM, Han S, Tang Y. A generalized physics-based calendar life model for Li-ion cells. *Electrochim Acta* 2020;348.
- [23] Bischof K, Flügel M, Hölzle M, Wohlfahrt-Mehrens M, Waldmann T. Aging Mechanism For Calendar Aging of Li-Ion Cells With Si/Graphite Anodes. *J Electrochem Soc* 2024;171(1):010510.
- [24] Madi Reddy SK, Shang W, White RE. Mathematical Model for SEI Growth under Open-Circuit Conditions. *J Electrochem Soc* 2022;169(9):090505.
- [25] Wang L, Ye W, Zhu Y, Yang F, Zhou Y. Optimal parameters selection of back propagation algorithm in the feedforward neural network. *Eng Anal Bound Elem* 2023;151:575–96.
- [26] Sapienza F, Bolibar J, Schäfer F, Groenke B, Pal A, Boussange V, et al. Differentiable Programming for Differential Equations: A Review. 2024.
- [27] Ma Y, Dixit V, Innes MJ, Guo X, Rackauckas C. A Comparison of Automatic Differentiation and Continuous Sensitivity Analysis for Derivatives of Differential Equation Solutions. In: 2021 IEEE high performance extreme computing conference. Institute of Electrical and Electronics Engineers Inc.; 2021.
- [28] Pannala S, Movahedi H, Garrick TR, Stefanopoulou AG, Siegel JB. Consistently Tuned Battery Lifetime Predictive Model of Capacity Loss, Resistance Increase, and Irreversible Thickness Growth. *J Electrochem Soc* 2024;171(1):010532.
- [29] Fan G, Lu D, Trimboli MS, Plett GL, Zhu C, Zhang X. Nondestructive diagnostics and quantification of battery aging under different degradation paths. *J Power Sources* 2023;557.
- [30] Kirkaldy N, Samieian MA, Offer GJ, Marinescu M, Patel Y. Lithium-Ion Battery Degradation: Measuring Rapid Loss of Active Silicon in Silicon-Graphite Composite Electrodes. *ACS Appl Energy Mater* 2022;5(11):13367–76.
- [31] Li R, O'Kane S, Huang J, Marinescu M, Offer GJ. A million cycles in a day: Enabling high-throughput computing of lithium-ion battery degradation with physics-based models. *J Power Sources* 2024;598.
- [32] Keskär NS, Mudigere D, Nocedal J, Smelyanskiy M, Tang PTP. On Large-Batch Training for Deep Learning: Generalization Gap and Sharp Minima. 2016.
- [33] Ai W, Kirkaldy N, Jiang Y, Offer G, Wang H, Wu B. A composite electrode model for lithium-ion batteries with silicon/graphite negative electrodes. *J Power Sources* 2022;527.
- [34] Chen C-H, Brosa Planella F, O'Regan K, Gastol D, Widanage WD, Kendrick E. Development of Experimental Techniques for Parameterization of Multi-scale Lithium-ion Battery Models. *J Electrochem Soc* 2020;167(8):080534.
- [35] O'Regan K, Planella FB, Widanage WD, Kendrick E. Thermal-electrochemical parameters of a high energy lithium-ion cylindrical battery. 2022, <http://dx.doi.org/10.26434/CHEMRXIV-2022-D2Q4N>.
- [36] Zilberman I, Sturm J, Jossen A. Reversible self-discharge and calendar aging of 18650 nickel-rich, silicon-graphite lithium-ion cells. *J Power Sources* 2019;425:217–26.
- [37] Maures M, Zhang Y, Martin C, Deléage JY, Vinassa JM, Briat O. Impact of temperature on calendar ageing of Lithium-ion battery using incremental capacity analysis. *Microelectron. Reliab.* 2019;100–101.
- [38] Olson JZ, López CM, Dickinson EJ. Differential Analysis of Galvanostatic Cycle Data from Li-Ion Batteries: Interpretative Insights and Graphical Heuristics. *Chem Mater* 2023;35(4):1487–513.
- [39] Bloom I, Jansen AN, Abraham DP, Knuth J, Jones SA, Battaglia VS, et al. Differential voltage analyses of high-power, lithium-ion cells 1. Technique and application. *J Power Sources* 2005;139(1–2):295–303.
- [40] Barai A, Uddin K, Dubarry M, Somerville L, McGordon A, Jennings P, et al. A comparison of methodologies for the non-invasive characterisation of commercial Li-ion cells. *Prog Energy Combust Sci* 2019;72:1–31.
- [41] Keil P, Jossen A. Calendar Aging of NCA Lithium-Ion Batteries Investigated by Differential Voltage Analysis and Coulomb Tracking. *J Electrochem Soc* 2017;164(1):A6066–74.
- [42] F. Rodrigues MT, Yang Z, Trask SE, Dunlop AR, Kim M, Dogan F, et al. Pouch cells with 15% silicon calendar-aged for 4 years. *J Power Sources* 2023;565.
- [43] McBrayer JD, Rodrigues MTF, Schulze MC, Abraham DP, Apblett CA, Bloom I, et al. Calendar aging of silicon-containing batteries. *Nature Energy* 2021;6(9):866–72.
- [44] Khaleghi Rahimian S, Forouzan MM, Han S, Tang Y. A generalized physics-based calendar life model for Li-ion cells. *Electrochim Acta* 2020;348.
- [45] Ekström H, Lindbergh G. A Model for Predicting Capacity Fade due to SEI Formation in a Commercial Graphite/LiFePO₄ Cell. *J Electrochem Soc* 2015;162(6):A1003–7.
- [46] Li R, O'Kane S, Marinescu M, Offer GJ. Modelling Solvent Consumption from SEI Layer Growth in Lithium-Ion Batteries. *J Electrochem Soc* 2022;169(6):060516.
- [47] Manmi K, Tüchel M, Kendrick E, Brosa Planella F. A Comparison of Standard SEI Growth Models in the Context of Battery Formation. *J Electrochem Soc* 2024;171(10):100530.
- [48] Cai J, Luo J, Wang S, Yang S. Feature selection in machine learning: A new perspective. *Neurocomputing* 2018;300:70–9.
- [49] Rackauckas C, Nie Q. DifferentialEquations.jl – A Performant and Feature-Rich Ecosystem for Solving Differential Equations in Julia. *J Open Res Softw* 2017;5(1):15.
- [50] Innes M, Edelman A, Fischer K, Rackauckas C, Saba E, Shah VB, et al. A Differentiable Programming System to Bridge Machine Learning and Scientific Computing. 2019.
- [51] Kingma DP, Ba J. Adam: A Method for Stochastic Optimization. 2014.

- [52] Lucu M, Martinez-Laserna E, Gandiaga I, Liu K, Camblong H, Widanage WD, et al. Data-driven nonparametric Li-ion battery ageing model aiming at learning from real operation data – Part A: Storage operation. *J Energy Storage* 2020;30.
- [53] Saldaña G, Martín JIS, Zamora I, Asensio FJ, Oñederra O, González-Pérez M. Empirical calendar ageing model for electric vehicles and energy storage systems batteries. *J Energy Storage* 2022;55.
- [54] Dufek EJ, Tanim TR, Chen B-R, Kim S. Battery calendar aging and machine learning. tech. rep..
- [55] Brunton SL, Proctor JL, Kutz JN. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proc Natl Acad Sci USA* 2016;113(15):3932–7.
- [56] Karniadakis GE, Kevrekidis IG, Lu L, Perdikaris P, Wang S, Yang L. Physics-informed machine learning *Nature reviews | Physics*.
- [57] Ying X. An Overview of Overfitting and its Solutions. *J Phys: Conf Ser* 2019;1168(2). Institute of Physics Publishing.